

MASTER'S THESIS

A REGRESSION MODEL FOR FLOOD WASTE PREDICTION IN ABIDJAN, COTE D'IVOIRE



by

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July 2024

Table of Contents

Table of Contents.....	2
List of Figures	4
List of Tables.....	4
Executive Summary	6
I. Introduction.....	8
II. Literature Review	9
III. The model.....	12
1. Methodology	12
1.1. Model formulation	12
1.2. Model features selection.....	12
1.3. Handling of cases of insufficient data	12
1.4. Quantification of waste amount: dependant variable for the model	12
1.5. Linear Regression Runs.....	18
2. Results	19
2.1. Model Features selection.....	19
2.2. Handling of case of insufficient data: estimation of remaining parameters.....	20
2.3. Quantification of waste amount: dependant variable for the model	21
2.4. Linear Regression Runs.....	32
3. Discussion: Further analysis, Interpretation	33
4. Conclusions and Limitations	37
IV. Environmental Impact Assessment.....	38
1. Methodology	38
1.1. Space required	38
1.2. Heavy metals.....	38
2. Results and Discussion	39
2.1. Space required	39
2.2. Heavy metals.....	40
3. Conclusions and Limitations	42
V. Conclusion.....	42
VI. References.....	43
VII. Appendices.....	46
1. Appendix A: The model	46
1.1. Method 1	51
1.2. Method 2.....	67
1.3. Method 3	69

1.4.	Method 4	73
1.5.	Method 5	75
1.6.	Models Runs.....	77
2.	Appendix B : EIA	102

List of Figures

Figure 1: Abidjan, Côte d'Ivoire on the Map	8
Figure 2: inondations in Abidjan (Dash, D. (2021) Abidjan inondations)	9
Figure 3: Usual flood waste composition	10
Figure 4: Split up of 2018-0185-CIV	20
Figure 5: Ratio of waste per household in Flood time vs in Normal time sorted in descending order	22
Figure 6: Flood waste per household in Bangkok	24
Figure 7: Flood Waste per person in ascending order – Korea 2012	25
Figure 8: Clustering based on Population density in flood affected regions - KOREA	27
Figure 9: Heavy metals in flood waste	42

List of Tables

Table 1: Table summary of 4 flood waste quantification methods.....	13
Table 2: List of desirable features.....	19
Table 3: Final list of features	19
Table 4: Clustering based on Ratio of waste per household in Flood time vs Normal time for Bangkok districts	21
Table 5: classification of social levels in Abidjan	22
Table 6: Adaptation from 1989 data to 2016	23
Table 7: Flood waste - Methods 1 & 2	23
Table 8: Flood waste per household Averages for each cluster // Table 9: Adaptation to Abidjan from Bangkok	24
Table 10: Adaptation from KOREA to Abidjan	25
Table 11: Adaptation from KOREA to Abidjan	26
Table 12: Flood waste – Method 4	27
Table 13: Main components of typical slum house in Abidjan	28
Table 14: Income ratio between different social classes	28
Table 15: Methodology for adaptation of Red Cross Reports Data.....	30
Table 16: Results after adaptation of Red Cross Reports Data.....	30
Table 17: Weight and quantity of Consumer Durables	31
Table 18: Total Consumer Durables Waste	31
Table 19: First model runs	32

Table 20: Results of 126 four-feature models runs.....	33
Table 21: Total Flood waste composition.....	36
Table 22: Comparative summary for all 5 methods.....	36
Table 23: Volume of Construction and Demolition Debris for CC houses	39
Table 24: Typical consumer durables for CC houses	39
Table 25: The heavy metals and their provenance.....	40
Table 26:Percentage of Metals from Total Waste	41
Table 27: Database Thailand	46
Table 28: Flood time vs normal time – Thailand.....	51
Table 29: Database KOREA.....	55
Table 30: Method 1 - Clustering of ratios.....	58
Table 31: Method 2 - Average waste per household	63
Table 32: Method 3.....	65
Table 33:	69
Table 34: Clustering KOREA.....	71
Table 35:	73
Table 36: Acronyms for damaged buildings.....	75
Table 37: Data collected and adapted from Red Cross reports.....	75
Table 38: Database Jumia; low, mid, and high-end typical items	76
Table 39: Runs with 3-feature models	101
Table 40: Percentage of each metal from all metals	102

Executive Summary

This work focuses on quantifying the waste generation after flood events in Abidjan, the capital city of Cote d'Ivoire. Its relevance lies in the major issue of recurring floods that affect this high-potential city every year during the rainy season. Literature review findings indicate that the principal causes of floods in places like Abidjan are a mix of natural events and anthropogenic actions: heavy rainfalls and river overflow exacerbated by inadequate infrastructure, city planning, and uncontrolled urbanization.

Actions have been taken to tackle this issue in West Africa, but no research on flood waste quantitative models was found for Abidjan and similar cities in neighbouring countries. Thus, in the first part of this study, a regression model using both a deterministic approach and a Bayesian Inference approach was developed in RStudio to predict the waste generated after any flood event in the city of Abidjan. Due to the extreme scarcity of relevant data, the objective was more on providing an example of a quantitative methodology that can be used to assess flood waste generation in Abidjan.

As the core part of building a deterministic regression model is to select the appropriate parameters, numerous parameters from previous research studies on flood waste in modelling in the world were gathered. Then, the feasibility of collecting data related to these parameters for the case of Cote d'Ivoire was assessed, resulting in a shortlist of features. The best parameters and the number of features needed were further evaluated using multiple runs with different combinations of parameters and certain performance metrics like VIF (Variance Inflation Factor). Additionally, because the flood waste (dependent variable of the model) generated in Abidjan after each historical flood event recorded was not registered, a special section was dedicated to its estimation using five different methods.

As a result, ten parameters were shortlisted: waste amount, population, area, urbanization ratio, number of households, amount of precipitation, GDP per area, hourly maximum rainfall, wind speed, and altitude. The five methods used for quantifying the waste amount after each historical event recorded in Abidjan provided a large range. Four of those methods were based on datasets from Thailand and Korea, and one of them was based on actual data from Abidjan. That data was derived from the Red Cross reports after their investigations on the Ivorian territory. Finally, the regression model with the highest adjusted R^2 of 0.75 had four features. Compared to other studies, the results for the metrics obtained are much higher, and that is probably due to slight overfitting issues. The posterior Bayesian updating results were derived from using the deterministic method results as prior for the mean and standard deviation of the distribution, i.e: the prior knowledge about the parameters.

Regarding the environmental assessment, the focus was on estimating the space required for temporary disposal of the flood waste and addressing the discharge of heavy metals contained in that waste. For the space required, the first section (The Model part) permitted finding the flood waste composition: construction and demolition debris, and consumer durables. Since that estimation was in terms of mass, the waste density was used to derive the total flood waste volume from the estimated mass. For the heavy metals, the lack of data was also an issue. Thus, the list of heavy metals to focus on was decided based on the Basel Convention regulations, and research articles about heavy metals usually found in the potable source of water in Abidjan.

The results were that the minimum volume range required as space for the flood waste was estimated to be 200-1400 m³ for a municipality. The principal heavy metals present in the flood waste were found to be copper, lead, steel, mercury, nickel, zinc, and aluminum. For the total amounts of flood waste every year, aluminum accounted for 95% of the heavy metals, and sensible metals like lead and mercury accounted respectively for 2ppm and 0.0021ppm. These small numbers compared to the limits advised by international standards indicated that there was no significant concern about the disposal of the heavy metals contained in the flood waste.

For future work, it will be mandatory to have much more access to real data from Abidjan. This could be achieved through surveys of local/government authorities and fieldwork. The existing data could also be enhanced by new data augmentation algorithms provided by the advancements in Artificial Intelligence. This would allow the use of advanced machine learning algorithms like neural networks for the prediction of flood waste in Abidjan. More importantly, incentives must be created in Cote d'Ivoire to generate more interest from the research community in this matter.

I. Introduction

Cote d'Ivoire is a multicultural country bordered by the Atlantic Ocean with a population of approximately 32 million people [1]. With a GDP of 87 billion USD in 2024, it ranked as the 9th largest economy in Africa and the leading economy among French-speaking West African countries. The nation possesses numerous assets that contribute to its promising future, including significant economic prospects in agriculture, minerals, and industry, as well as a youthful demographic, with individuals under 15 years old comprising 38.2% of the total population[2]. However, [3] several risks threaten these prospects, one of which is the recurring issue of flooding. Due to its geographical location and economic conditions, Cote d'Ivoire has experienced numerous flood events over the decades. The primary causes of these floods include inadequate infrastructure and city planning, compounded by a significant rainy season from June to August. These events have resulted in numerous fatalities, injuries, and displacements. Additionally, climate change has exacerbated the situation in recent years, with predictions indicating increasingly devastating effects in the future.



Figure 1: Abidjan, Côte d'Ivoire on the Map

The focus of this study will be Abidjan, the capital city of Cote d'Ivoire. It consists of 6 million people living across its 10 municipalities. Abidjan exhibits a diverse mix of social classes, including the wealthy, upper middle class, middle class, and very poor. It has witnessed a significant economic boom which has led to a surge in the construction of many houses with uncontrolled planning. All residents from any social class are more or less affected by the flooding issue, but the poorest are the most vulnerable. They often live in inadequate housing, such as slums, and lack the means to recover from flood events, which occur annually during the rainy season.

In general, there is a lack of research on quantitative models to predict the waste generated by flood events. Although multiple studies focus on large-scale disasters with tremendous damages [4], few have explored flood waste prediction using linear regression, logistic regression, and more complex machine learning models. However, no research on predictive models for flood waste in the West African context was found. This highlights the critical need for comprehensive and predictive models in West Africa.



Figure 2: inondations in Abidjan (Dash, D. (2021) Abidjan inondations)

Thus, in this study, a regression model is developed to predict the waste generated after any flood event in the city of Abidjan. The goal is not to find exact numbers, but more about providing an example of quantitative methodology that can be used to assess the flood waste generation in Abidjan. The first part involves deriving one model from developing and testing multiple deterministic regression inspired by other models used to assess flood impacts, including environmental and socio-economic factors [6]. From the deterministic regression models, a Bayesian Inference approach is used to test the prediction by incorporating more uncertainty about the parameters. The second part of the study focuses on conducting an environmental impact assessment of the flood waste estimation. This assessment will address the discharge of heavy metals contained in the waste and the space required for a temporary disposal site.

II. Literature Review

C.N Egbinola et al [7] chose Ibadan, Nigeria as a case study for flood management in cities of developing countries. They identified the natural causes of floods to be heavy rainfall, river overflow, and climatic changes. These natural causes are exacerbated by human activities such as deforestation and rapid urbanization, compounded by poor planning. They found that mismanagement, lack of continuity in governance, and policy are significant barriers to effective flood risk management in Ibadan. Nevertheless, they recommended the intensification of non-structural flood management strategies, as small-scale flood management remedies such as dredging and clearing channels and bridge thoroughways have proven efficient.

Kablan et al.[8] showed that high-risk flood areas in Abidjan include districts with poor drainage systems and those located in low-lying regions. The main areas identified are Cocody, due to its topography and population density, and Attécoubé, because of its location and infrastructure challenges. The researchers adapted the MOVE framework (Methods for the

Improvement of Vulnerability Assessment in Europe) to the Ivorian context to assess social vulnerability to floods. They considered various factors such as exposure (geographic extent and population density in flood-prone areas), susceptibility (predisposition to harm based on socio-economic conditions and infrastructure fragility), and resilience (capacity to cope and recover from floods). This framework was successfully applied in urban Côte d'Ivoire, including Abidjan, to identify and map vulnerable areas, thereby guiding local authorities in emergency response planning and infrastructure improvement.

Regarding the use of quantitative methods, Fotovatikhah F. et al. [9] extensively reviewed research on computational intelligence methods as basis to big flood management. They pointed out that artificial intelligence and computational intelligence methods are generally can be categorized as single and hybrid approaches and are extensively applied to early flood event detection. These methods enable city authorities to provide quick and efficient responses in post-disaster scenarios, with ensemble computational intelligence methods being identified as the most efficient. Marchesini G. et al. [10] focused their quantitative models review on the prediction of disaster waste generated. Due to the limited available research on this subject, they examined approximately 20 studies related to different disasters worldwide, including floods, hurricanes, earthquakes, and tsunamis. The main types of waste identified were Construction and Demolition Debris, Vegetation Waste, and Household Good Wastes. Most studies focused on quantifying one or two types of these waste categories. Other wste components can be seen in Figure 3 below.

Construction and Demolition debris	Vegetative debris // Soil, Mud and Sand	Putrescible wastes
Industrial and toxic chemicals	Electronics and white goods	Hazardous waste (insecticides, pesticides)
Healthcare Waste	Human and animal bodies	Emergency assistance waste

Figure 3: Usual flood waste composition

In addition, three types of methods were used for waste quantity estimation: historical data methods, [national/regional/local] databased methods, and imaging methods. Historical data & databased methods are mainly used before the disaster for anticipation of waste planning, and imaging methods serve as primordial tools for emergency response and recovery phase planning.

Soro G. et al. [11] conducted one of the very few studies on heavy metals in Abidjan. They analyzed the sediments of the Ebrie Lagoon in Côte d'Ivoire to identify the extent of heavy metal pollution. Their focus was on determining the sources and levels of chromium, copper,

manganese, and zinc in the collected sediments. The results indicated that the levels of these metals were not high enough to raise significant concerns, despite the presence of industrial activities.

III. The model

1. Methodology

All datapoints for this study regarding flood casualties in Abidjan were derived from the International Disaster Database EM-DAT [13]. Due to the initial data being difficult to exploit directly, several adaptations were made. An example of the adaptations can be seen in Figure 4.

1.1. Model formulation

- Deterministic and Bayesian Linear regression from Mathematics for Machine Learning [14]
- lm function and rstan library in RStudio

1.2. Model features selection

To reiterate, the goal of the first part of this research is to create a regression model to predict the waste generated after any flood event in Abidjan. The core aspect of building a regression model is selecting the appropriate parameters. To do this, Numerous parameters from previous research studies on flood waste modeling were gathered. Subsequently, the feasibility of collecting data related to these parameters for Cote d'Ivoire was assessed. Based on this assessment, the parameters were narrowed down according to availability, resulting in a shortlist of features for the model. The data needed for the model was primarily collected from official government websites of Cote d'Ivoire, reputable research papers on the topic, and websites of organizations such as the World Bank and IMF. The relevance of the parameters was further evaluated by performing multiple regression analyses using different sets of parameters and comparing the accuracy of the coefficient of determination and other metrics.

1.3. Handling of cases of insufficient data

When the parameters necessary to build the model lacked relevant data for Abidjan, estimates were made. These estimations were based on information from external research papers to closely approximate the real scenario. For instance, as there was no data on the GDP per area in each municipality of Abidjan, the method to approximate the GDP per area relied on economic activities, population density, and other indicators of quality of life in each municipality.

1.4. Quantification of waste amount: dependant variable for the model

In the specific case of the amount of waste generated after each flood event, which is the dependant variable of the regression model, no data was initially available. Therefore, a special methodology (Table 1) was developed to estimate this quantity, employing five methods derived from the work of researchers in other countries.

The fifth method served as the primary basis for the estimation, while the first four methods served to provide a range of waste quantity estimates. It was chosen because it is mostly based on data relevant to real-life conditions in Abidjan.

Table 1: Table summary of 4 flood waste quantification methods

	Method 1	Method 2	Method 3	Method 4
Similarities and differences	same Flood dataset from Thailand		Use same flood dataset from Korea	
	Clustering methods and one average value assigned to each cluster			
	3 clusters	5 clusters	5 clusters	3 clusters
Key parameter	Ratio total waste in flood time to total waste in normal time	Per household total waste in flood time	Flood waste per person	Flood waste per km ²
Main Assumptions	Flood duration considered		Flood duration not considered	
	Steady increase in waste generation from 1989 to 2016 in Abidjan Richer areas are relatively less prone to flood waste generation	Place more exposed to floods events in Abidjan have higher flood waste generation	The whole population of each municipality in Korea generated flood waste	Locations with similar densities have similar rate of flood waste generation

Furthermore, for each of those methods, many assumptions (presented in following sub-sections) were made; otherwise, obtaining results would have been impossible due to the nature of the project.

Equation 1 to Equation 5: below were used for the quantifications for all five methods. They were taken from previous research studies [10].

Method 1: Bangkok-based

Equation 1:

$$flood\ waste = \left(\frac{average\ waste_{flood\ time\ BANGKOK}}{average\ waste_{normal\ time\ BANGKOK}} \right) * daily\ waste_{Abidjan\ normal\ time} * days\ of\ flooding$$

This method is based on data available for a series of flood events recorded [15] from December 2011 to January 2012 in Bangkok, Thailand. The fundamental assumption is that the ratio of waste generated in flood time to normal time in Bangkok during those flood events is approximately the same in Abidjan. There were 45 locations recorded for Bangkok, but Abidjan had 7. Therefore, the same number of clusters are created for Abidjan and Bangkok to match them.

- Clustering for Abidjan: The 7 municipalities are separated in three clusters based on three social classes [15] : low class, middle class, and upper class.
- Clustering for Bangkok: The ratios are sorted in ascending order and also put into three clusters based on the trend of the plotted curve.

The three clusters created are matched to one another following this logic: The higher the quality of life (Abidjan), the lower the flood waste generation (Bangkok).

Some of the remaining assumptions for the estimation are the following:

- Waste generation in normal time is proportional to flood waste
- Waste generation per capita in normal time for households in Abidjan increased at the same rate, every year, from 1989 to 2016.
- Clusters that were created for Bangkok have the same waste generation rate as the clusters created for Abidjan.
- The rate of generation of flood waste in Bangkok in 2011-2012 is similar to the one in Abidjan in 2015.

Method 2: Bangkok-based

Using the same data from Bangkok as in Method 1, the difference here is that the flood waste generated per household for each district in Bangkok is the key element. Here as well, clusters are created and matched assuming that the places more exposed to floods in Abidjan have higher flood waste generation.

Equation 2:

$$flood\ waste = \sum_i (flood\ waste\ per\ household_{Bangkok} * number\ of\ households)$$

Some of the remaining important assumptions are the following:

- The more exposed places to flooding in Abidjan have a higher risk of waste generation. (Have to be critically analysed with programming by testing different outcomes if we change the exposed locations)
- Number of people per household do not differ from 2015 to 2021

Method 3: Korea-based

This method for estimating the amount of flood waste is based on data collected from studies on Korea flood modelling [16] [17] [18].

Equation 3:

$$flood\ waste = \sum_i (flood\ waste\ per\ person_{KOREA} * population_i)$$

i = population category

It follows a similar clustering approach to the first two methods. The focus is put on the recorded events in 2012 for the entire country (Appendix-Table 34). The fundamental assumption is that the generation of flood waste in Korea in 2012 is approximately the same as that of Abidjan between 2015 and 2021. There were about 37 municipalities recorded for Korea, but 7 in Abidjan for that study. Hence, the same number of clusters are created for Abidjan and Bangkok to match them.

- Clustering for Abidjan: The 7 municipalities are separated into five clusters based on the probability of flood waste generation: low class, middle class, and upper class.
- Clustering for Bangkok: The ratios are sorted in ascending order and also put into five clusters based on the trend of the plotted curve.

The five clusters created are matched following this logic: The higher the quality of life (Abidjan), the lower the flood waste generation (Korea).

Some of the remaining assumptions for the estimation are the following:

- The whole population of each municipality generated flood waste
- The yearly increase of the rate of flood waste per capita in Abidjan from 2015 to 2021 follows the same rate as that of the normal municipal waste used for Method 1.

Method 4: Korea-based

Using the same data from Korea as in Method 3, the difference here is that the clusters are based on waste generated per population density (Table 35). The estimation principally assumes that clusters of locations with same density have the same generation of flood waste. The classification of the different densities for the clustering was based on relatively to other places within the same location (Korea or Abidjan).

Equation 4

$$flood\ waste = \sum_i (flood\ waste\ per\ cluster\ of\ density_{KOREA} * density\ of\ cluster\ i)$$

$$i = population\ category$$

As an example, let's take the case of municipalities in Korea which a high density (relatively to other places in Korea, corresponds to more than 1100 people/km²), and the case of municipalities in Abidjan with also a high density (relatively to other places in Abidjan, corresponds to more than 17,000 people/km²); thus, based on the assumption stated earlier, the cluster of high-density municipalities in Korea have a close flood waste generation rate as the high-density municipalities in Abidjan. After that, a single value for waste generated per density is attributed to the whole cluster by taking the average (see Table 11).

Method 5

The fifth method serves as the primary basis for the estimation. It uses data that is most relevant to real-life conditions in Abidjan, instead of adapting data from Thailand and Korea. Following Equation 5, it consists in finding the weight of construction & demolition debris and consumer durables which turn into flood waste. To do so, due to the lack of information available on all

the building types for the municipalities (residential, commercial, industrial, etc), the affected buildings are assumed to be mostly houses.

- The houses are classified into three categories: completely destroyed (CC), partially destroyed (MC), and floor level inundation (FL). N_i , the number of those damaged buildings (see Equation 5), was estimated for each flood incident based on an extrapolation of data gathered by the Red Cross (Table 37).
- The weight of a typical house in a slum will be estimated and serves as the basis to waste from find Construction & Demolition Debris in CC and MC.

Equation 5:

$$W_D + W = \sum C_i N_i + h_i * \sum c_i * m_i$$

where:

i = population category (social class?)

W_D = construction and demolition debris

W = household waste

C_i = per unit generation of debris for building damages

N_i = number of buildings in damage category

h_i = number of completely destroyed houses

c_i = quantity of consumer durables i possessed per household

m_i : mass per unit of the major consumer durables i

Some of the main assumptions used for the estimation are the following:

- Total flood waste is majoritarilly made of consumer durables and construction and demolition debris
- All houses in slums are completely destroyed and turned into flood waste
- The weight of a slum house is proportional to the weight of a house outside a slum
- All houses in slums have the same furniture
- The ratio of waste generated between CC, MC, and FL are proportional
- For the estimation of casualties by the Red Cross organisation, the casualties for the municipalities ignored were estimated by assuming that all of them had been

affected to the same extent/with the same gravity. Nevertheless, some were adapted by following the pattern for the previous years.

❖ Method 5: 1st Part: Estimation of Construction and Demolition Debris (W_D)

In that study, the waste identified as Construction and Demolition Debris refer to any part of one house that is destroyed, excluding items brought by the inhabitants such as furniture, beds, chairs, consumer durables, clothes, etc.

The initial data provided by the Red Cross [19] [20] [21] [22] regarding the floods and numerous flood waste incidents in Abidjan was crucial for information of flood casualties. Those reports are the only publicly available resources that contain information that can be used for the purpose of identifying households affected by floods in Cote d'Ivoire. Specifically, it was used to estimate the total number of houses completely (CC) and moderately damaged (MC) by flood incidents. Subsequently, the weight of one house in a slum area was estimated using information about the foundation, walls, roof, flooring, furniture, appliances, clothes, and other typical household items found in a slum. This estimation was based on the density of the primary materials used to construct these elements.

To estimate the weight of middle-class (MC) houses and upper-class (CC) houses, the assumption was that the weight ratio of MC houses to CC houses is the same as the income ratio of MC households to CC households. For example, in 2018, the income for middle-class people was four times higher than that for poor people. We applied the same ratio to estimate how many times heavier the weight of a middle-income house is compared to a slum house. Additionally, a factor was added because houses in the CC and flood-affected (FL) categories are not completely destroyed. We assumed that 30 percent of each MC house were partially destroyed and one percent of each FL house was partially destroyed. Moreover, the data from the Red Cross did not provide specific details for each municipality in Abidjan but only for a few of them. Therefore, more adaptations were made, using ratios and clusters, to estimate the casualties for each municipality.

❖ Method 5: 2nd Part: Estimation of Consumer Durables (W)

This part focuses on the estimation of consumer durables. The initial method [10] from the literature only considered consumer durables from CC houses, but in our study, we also included consumer durables that turned into waste from MC houses and FL houses. The key parameters here are the unit weight (in kg) of each item and how many of them one CC household possesses them on average. The same Red Cross data for estimating the total number of affected households was used here again.

The estimation follows the main assumption that typical consumer durables are furniture and appliances. A database of low-end, mid-end, and high-end furniture was created by researching prices, weights, and materials used for these items on the popular website Jumia [23], the Amazon equivalent of Cote d'Ivoire.

In reality, CC households use second-hand or discarded items, but relevant data about this was not found. To compensate, the average was taken. For example, among types of beds found on Jumia, low-end beds usually weigh 4 kg, mid-end beds 18 kg, and high-end beds 25 kg. The average weight of 16 kg was chosen as the typical bed weight for slum houses.

Research papers on e-waste in West Africa were also reviewed to understand the types of consumer durables and materials used.

Regarding the remaining assumptions the main ones are:

- Some electronic devices such as FM radios and mobile phones become lighter over time because of technology advancements. This was not taken into account in the estimations.
- The typical consumer durables in all slums are the same whatever the municipality.

1.5. Linear Regression Runs

As said earlier, the goal of the first part of this research work was to create a regression model to predict the waste generated after any flood event in Abidjan. After completing the features selection, many models tested in previous research studies on flood waste modelling were tried on the datasets created for Abidjan municipalities.

Data preparation

The final dataset consists of the database created from the previous parts of this section, with the explained variable being the waste quantity estimated in Method 5. Because of the different scales for each parameter, normalization was done to keep all the values within the same range to facilitate the regression. Additional analysis like correlation and summary statistics were also conducted.

First model runs

The first model runs involved studies from Taiwan and Korea. In the study from Taiwan, a linear regression model was compared with an exponential model. For Korea, the same was done, but with a different number of features. In our case, we tested the same features used in each of those studies on the Abidjan datasets.

After running those models, we then tested different combinations of all the features that were shortlisted in 1.2. Model features selection. We tried combinations of four from the nine features shortlisted on the dataset. This resulted in about $\binom{9}{4} = 126$ model runs. After that, the top five models in terms of adjusted R-squared values were recorded and considered the most efficient four-feature models.

Bayesian updating

Bayesian updating is based on Bayesian inference and is a different approach from deterministic statistics. Here, the true value of a parameter of the regression model is considered as a random variable instead of a single (deterministic) value. This random variable's probability distribution is estimated by consecutive corrections of distribution from the probability of observed events under a designed prior belief [24]. Adding the Bayesian updating approach for this study was inspired by the work of [3]. They performed it to evaluate the performance and assess effectiveness of Bayesian approaches in flood waste estimations. The Rstanarm package [25] in Rstudio were used to perform Bayesian updating after the results of the first model runs part.

2. Results

2.1. Model Features selection

The parameters initially selected as potential features for the model based on the literature review are a mix of pre-disaster (regional characteristics) and post-disaster parameters (damages).

Table 2: List of desirable features

• Waste amount • Population • Area • Urbanization ratio • Residential land area • Slump land area • Commercial land area • Industrial land area • Landfill sites • Households • Amount of precipitation • Flood duration	• Length of rainy season • Temperature • Income per capita • GDP per area • Area of the flooded region • Hourly maximum rainfall • Wind speed • Damaged buildings • Damaged croplands • Damaged roads • Damaged rivers infrastructure • Altitude
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Table 3: Final list of features

• Waste amount • Population density • Flood duration • Urbanization ratio • Number of Households	• Amount of precipitation • GDP per area • Hourly maximum rainfall • Wind speed • Altitude
---	---

The only official website database for flood data in Cote d'Ivoire [13] was unexploitable for the objectives of this study. There were 13 rows of information with a lot of missing data. For example, the case “2010-0255-CIV” was reported as a single event of multiple floods in Bondoukou city, Abidjan district. The focus of this study is Abidjan municipalities, but there was no available information found from other sources on the damages for the municipalities specifically. Therefore, the “2010-0255-CIV” was not considered in our research. Furthermore, since the rows contained multiple locations simultaneously, each of the rows that we kept was split up. Taking the example of “2018-0185-CIV”, **as illustrated through Figure 4**, all these locations were reported as a single event on June 18th, 2018: Abobo, Adjamé, Attécoubé, Cocody, Koumassi, Yopougon districts, Tiassalé, Toulepleu, San Pedro. Consequently, the row was split up and only the municipalities of Abidjan were kept as multiple events happening on this June 18th. After that split up, data for the model features selected were gathered from different sources.

DisNo.	...	Location	...	Start Month	Start Day
2018-0185-CIV	...	Abobo, Adjamé, Attécoubé, Cocody, Koumassi, Yopougon districts (Abidjan); Tiassalé, Toulepleu, San Pedro	...	6	18

Year	Location
2018, June 18th	Abobo
	Adjame
	Attecoubé
	Cocody
	Koumassi
	Yopougon

Figure 4: Split up of 2018-0185-CIV

From all the elements of Table 2, only the Household size, Population, and Area for each locality in Abidjan could be found on official government websites, or highly trusted sources. Thus, to have a minimum of data to work with, some adaptations – described in the following section 2.2 – were made to estimate a part of the other parameters (Table 3).

2.2. Handling of case of insufficient data: estimation of remaining parameters

Because of no official record of urbanization ratio, precipitation, GDP per municipality, and wind speed for the municipalities in Abidjan, data were created using different methods.

Urbanization ratio (UR)

The municipalities were categorized into four clusters based on relative economic activity and residential habitations. They were respectively assigned Urbanization Ratio values ranging from 20 to 85. As an example, the municipality of Cocody has high economic activity and multiple mid- to upper-class residential places. On the other end, Attecoubé consists of small

businesses and a populous low- to mid-class residential population. Therefore, Cocody was put in the Cluster 1 (UR = 85), while Attécoube was put in the Cluster 4 (UR = 20).

Amount of precipitation, rainfall, windspeed, GDP per Area

The data was taken from the Weather and Climate website [26] to insure consistency in the values. For the GDP per area, a method similar to the one used for UR was used. In this case, the clustering was made only based on economic activities.

2.3. Quantification of waste amount: dependant variable for the model

A comparative summary of the results for all 5 methods can be seen in Table 22.

Method 1

As shown in Table 4, the ratio of waste in flood time to waste in normal time in Bangkok districts ranged from 1 to 3.15. Figure 5 shows how some districts stand out from others in terms of flood waste generated. Grouping them into three clusters seemed the best regarding the trend of the plot on Figure 5. The average waste ratio for the three clusters were 1.09, 1.25, and 1.83.

Table 4: Clustering based on Ratio of waste per household in Flood time vs Normal time for Bangkok districts

Number	District	Ratio normal/total	Clusters	Average Ratio normal/total
1	Don Mueang	3.15	TIER 3	1.83
2	Sai Mai	1.81		
3	Lak Si	1.50		
4	Phra Nakhon	1.37		
5	Khan Na Yao	1.31		
6	Bang Khae	1.27	TIER 2	1.25
7	Bang Khen	1.26		
8	Nong Khaem	1.23		
9	Bang Phlat	1.23		
10	Pathum Wan	1.19		
11	Thawi Watthana	1.19	TIER 1	1.09
12	Huai Khwang	1.18		
...	...	...		
...	...	...		
41	Prawet	1.01		
42	Bang Bon	1		

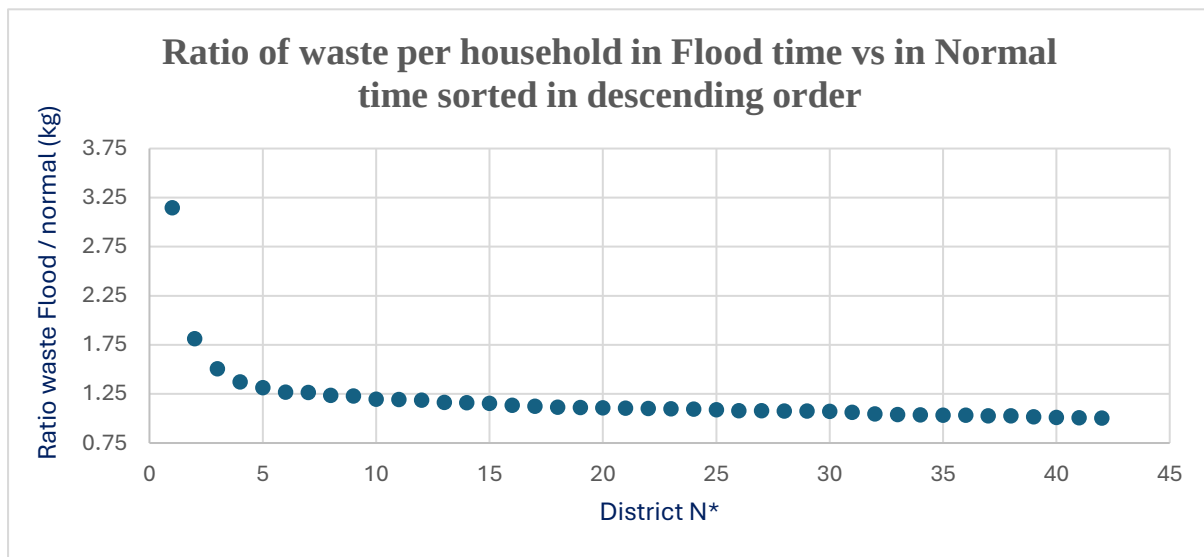


Figure 5:Ratio of waste per household in Flood time vs in Normal time sorted in descending order

The three clusters for Abidjan municipalities can be seen in Table 5. The Tier 1, being the upper class, was matched with the least flood generating Bangkok Tier 1.

Table 5: classification of social levels in Abidjan

TIER 3	TIER 2	TIER 1
Low class	Middle Class	Upper class
0.5 kg/day/person	0.8 kg/day/person	1 kg/day/person
Abobo Attecoubé Port Bouet Koumassi Yopougon	Adjamé	Cocody Plateau

On the assumption that the rate of generation of flood waste in Bangkok in 2011-2012 was similar to the one in Abidjan in 2015, the flood events for the following years were derived from that baseline. In more detail:

- Step 1: The waste generated in normal time from 2015 to 2021 in Abidjan municipalities
 - The waste generated every year in Abidjan was not available. However, according to a 1989 study [27], the average waste generated per household in the municipalities was found to be 0.5, 0.8, and 1kg/day/person respectively for lower, middle, and upper classes.
 - Also, according to the World Bank - What a waste 2.0 [28] , Abidjan had 0.67 kg of waste/day generated in 2016.

Thus, from that information, a steady increase in waste was assumed to be generated every year from 1989 to 2016. By using the compound interest formula (Table 6), it was found to be 0.64%.

Similarly, the population for each municipality for every year was computed using the compound interest formula with a yearly increase of 3%. As a recall, Equation 1 used to estimate the waste requires the *population* and the *normal waste generated* for a municipality every year.

- Step 2: Using the matching strategy described earlier, the waste for each flood event recorded has been estimated and can be seen in Table 7. They range from 65 thousand to 6.5 million kg.

Table 6: Adaptation from 1989 data to 2016

	TIER 1 (kg/day/person)	TIER 2 (kg/day/person)	TIER 3 (kg/day/person)	Daily Total Waste Average	Rate Increase Per Year
1989	0.50	0.80	1.00	0.56	0.0064
2016	0.59	0.94	1.18	0.67	

Table 7: Flood waste - Methods 1 & 2

Year	Location	Population	Normal waste generated (kg)	Flooding Days	TOTAL FLOOD Waste (kg)	
					METHOD 1	METHOD 2
2015, 7th June	Adjame	285,492	269,326	14	934,366	662,179
	Attécoubé	262,246	154,623	14	1,795,261	2,035,786
2017, 10th June	Abobo	1,190,573	710,932	11	6,485,555	6,959,228
	Koumassi	366,307	218,735	11	1,995,431	747,034
	Yopougon	1,395,871	833,522	11	7,603,901	2,656,911
2018, 18th June	Abobo	1,226,290	736,917	1	611,146	651,637
	Adjame	311,964	299,951	1	74,330	51,684
	Attécoubé	286,563	172,205	1	142,814	48,508
	Cocody	633,812	761,756	1	65,240	43,645
	Koumassi	377,296	226,730	1	188,033	39,576
	Yopougon	1,437,747	863,989	1	716,530	248,783
2020, 15th June	Abobo	1,300,971	791,773	2	1,313,279	1,382,643
	Adjame	330,963	322,279	2	159,725	109,664
	Anyama	378,245	230,200	2	381,823	401,990
	Cocody	672,411	818,460	2	140,192	92,607
	Port Bouet	600,772	365,631	2	606,456	222,762
	Yopougon	1,525,306	928,303	2	1,539,736	527,869

Method 2

As said earlier, this method is very close to the previous one and uses the same data from Bangkok. Here, the **flood waste generated per household** for each district (instead of the ratio) is the key element, and five clusters (Table 8) were derived from the trend of

Figure 6. The number of people per household in Abidjan was taken to be constant at 4.8 from 2015 to 2021. The amount of waste per household for each location

can be seen in the Appendix - [Method 2](#) .From cluster/tier 1 to 5, the average flood waste per household ranges from 5 to 153 kg for the two months of flood events. After adaptation to the actual flood duration for each municipality in Abidjan, the total flood wastes were calculated. As can be seen in Table 7, they range from 39 thousands kg to 2 million kg.

Table 8: Flood waste per household Averages for each cluster // Table 9: Adaptation to Abidjan from Bangkok

Tiers (Bangkok)	Average Flood Waste per household (kg)	Tiers (Abidjan)	Location (kg)	Flood Waste Per Household (kg)
Tier 5	5.12	Tier 1	ANYAMA	153.04
			ABOBO	153.04
		Tier 2	ADJAME	46.72
			PORT BOUET	46.72
			ATTECOUBE	46.72
			YOPOUGON	46.72
Tier 4	16.94	Tier 3	KOUMASSI	26.43
Tier 3	26.43		TREICHVILLE	26.43
Tier 2	46.72	Tier 4	COCODY	16.94
Tier 1	153.04	Tier 5	PLATEAU	5.12

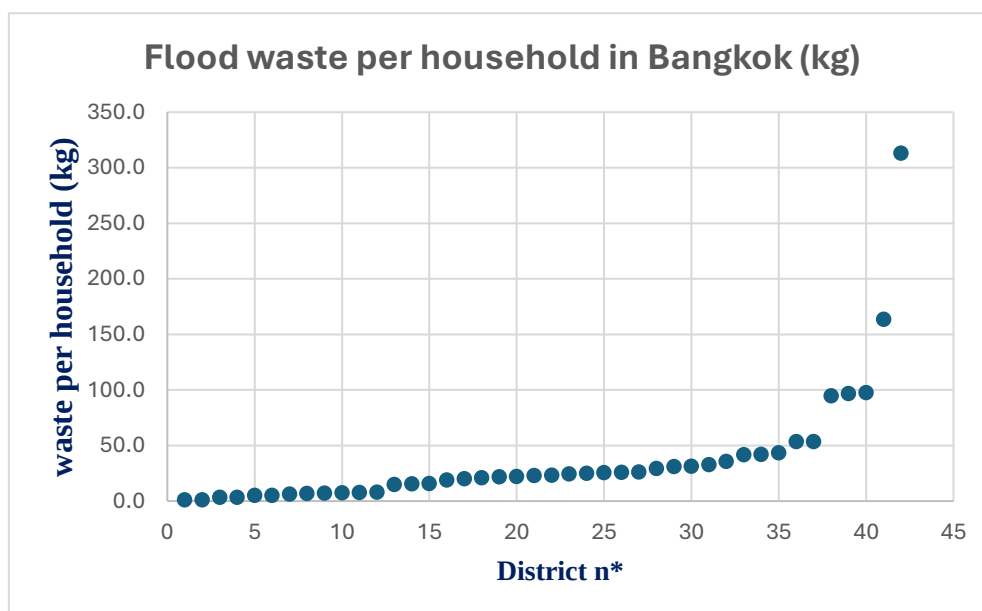


Figure 6: Flood waste per household in Bangkok

Method 3

From the 37 municipalities in Korea and the 7 municipalities in Abidjan that were focused on, 5 clusters (tiers) were created. As can be seen in Table 34, cluster 1,2,3,4, and 5 have, respectively, an average of 50, 20, 7, 3, and 0.5 kg of waste/person. As explained in the methodology, those numbers have been matched to the year 2015 in Abidjan, and adapted to the following years by assuming a yearly increase of 0.64%. Finally, the predicted amount of flood waste generated in Table 7, Abidjan municipalities ranges from 2 million to 64 million.

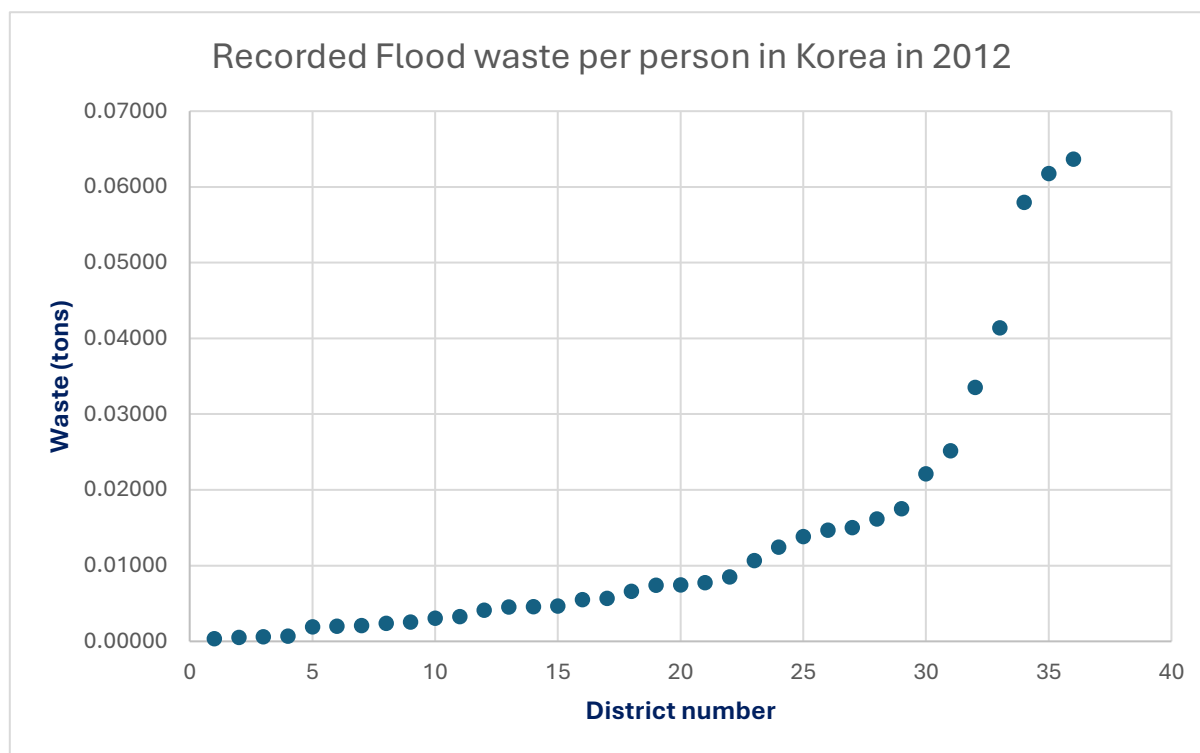


Figure 7: Flood Waste per person in ascending order – Korea 2012

Table 10: Adaptation from KOREA to Abidjan

		Flood waste (kg) / person from 2015 to 2020 using 1989 paper yearly rate of increase in normal waste			
	Location	2015	2017	2018	2020
TIER 1	ANYAMA	51.66	52.28	52.59	53.23
	ABOBO				
TIER 2	ADJAME	16.40	16.59	16.69	16.90
	PORT BOUET				
	ATTECOUBE				
	YOPOUGON				
TIER 3	KOUMASSI	6.69	6.77	6.81	6.89
	TREICHVILLE				
TIER 4	COCODY	3.04	3.08	3.09	3.13
TIER 5	PLATEAU	0.54	0.55	0.55	0.56
(MAYBE NOT NEEDED!!) Yearly % Rate of increase waste = 0.006360638					

Method 4

Three clusters were created from the selected seven municipalities in Abidjan and thirty-seven municipalities in Korea. They were all separated relatively from the other locations within the same country, explaining why the threshold for clusters in Korea is different from the one in Cote d'Ivoire (Abidjan).

- The “Not dense” locations in Korea have a density between 42 and 506 people per km² while the “Not dense” locations in Abidjan have a density lower than 5000 people per km².
- The “Dense” locations in Korea have a density between 605 and 1005 people per km² while the “Dense” locations in Abidjan have a density between 5,000 and 8,500 people per km².
- The “Very dense” locations in Korea have a density higher than 1,100 people per km² while the “Very dense” locations in Abidjan have a density higher than 17,000 people per km².

The average waste for each cluster can be seen in Table 11, and the final flood waste prediction for Abidjan municipalities is in Table 12.

Table 11: Adaptation from KOREA to Abidjan

Location	Population	Density	Tiers	Classification	Matching with KOREA Waste/density average (kg/person/km ²)
Anyama	389,592	760	1	Not dense	5.84
Cocody	692,583	5,299	2	Dense	2.61
Attecoubé	313,135	5,680			
Port Bouet	618,795	8,144			
Yopougon	1,571,065	8,501			
Abobo	1,340,000	17,192	3	Very dense	0.60
Koumassi	412,282	19,098			
Adjame	340,892	24,485			

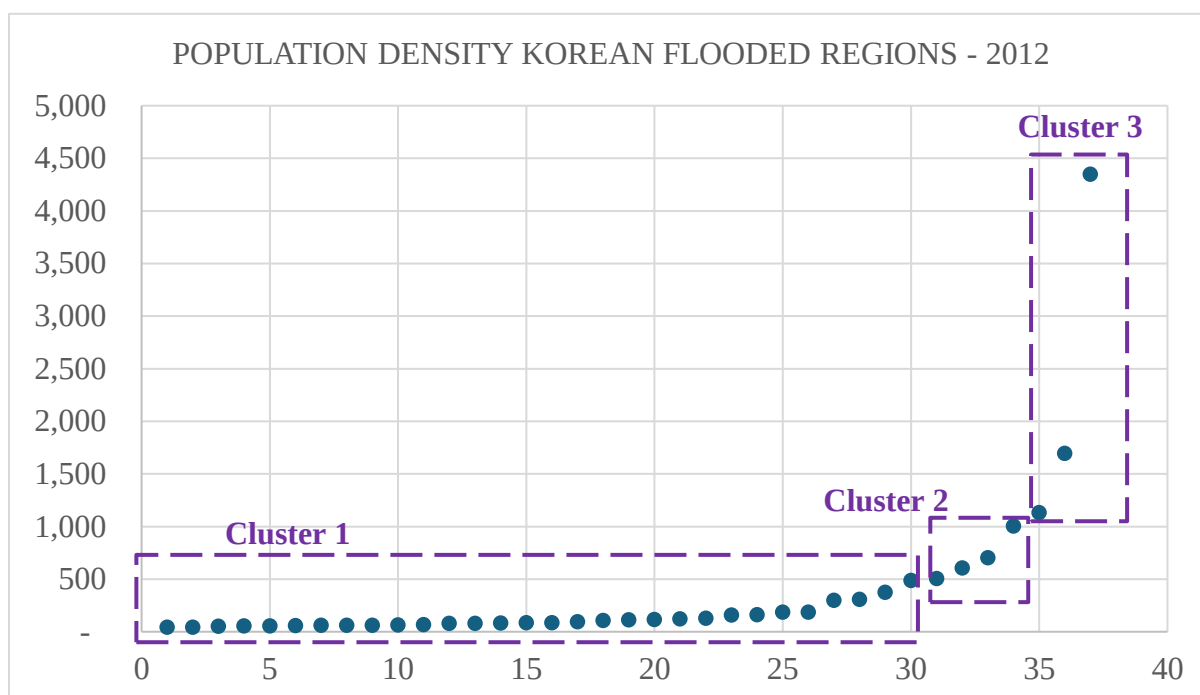


Figure 8: Clustering based on Population density in flood affected regions - KOREA

Table 12: Flood waste – Method 4

Year	Location	Population Density	Classification	Flood Waste (Kg)
2015	Adjame	24,485	Very dense	14,716,414
	Attecoubé	5,680	Dense	14,841,450
2017	Abobo	17,192	Very dense	10,333,399
	Koumassi	19,098	Very dense	11,478,997
	Yopougon	8,501	Dense	22,212,633
2018	Abobo	17,708	Very dense	10,643,401
	Adjame	26,755	Very dense	16,081,023
	Attecoubé	6,207	Dense	16,217,653
	Cocody	5,299	Dense	13,847,039
	Koumassi	19,671	Very dense	11,823,367
	Yopougon	8,756	Dense	22,879,012
2020	Abobo	18,787	Very dense	11,291,584
	Adjame	28,384	Very dense	17,060,357
	Anyama	760	Not dense	4,432,724
	Cocody	5,622	Dense	14,690,324
	Port Bouet	8,144	Dense	21,279,319
	Yopougon	9,289	Dense	24,272,344

Method 5

❖ 1st Part: Estimation of Construction and Demolition Debris (W_D)

Based on the initial data provided by the Red Cross [19] [20] [21] [22] numerous flood waste incidents in Abidjan was summarized in Table 37. According to the reports, the total number of houses affected ranged from 2,000 to 5,000 every year since 2018 and CC houses usually accounted for 10% of them.

For this study, the items decided as the main components of a CC house (typical house in a slum) were the foundation, walls, roof, flooring, furniture, appliances, and clothes. The building materials for houses in slums were identified as mud or earth for the building material, Portland cement as the binding material, and aluminum sheets for the roof (Table 13).

Table 13: Main components of typical slum house in Abidjan

Main components	Material	Density (kg/m ³)
Building material	Mud/Earth (free)	1890
Binding material	Portland cement	1506
Roof	Aluminum sheet	2739
liquids/additives	water	1000

The estimation of the weight of one house computed from the density of the materials, after taking a 20% margin of error, resulted in a weight range of 12,000 -18,000 kilograms. **In other words, when one CC house is destroyed, the construction and demolition debris are between 12 tons and 18 tons.**

The income ratios between the different social classes that were used for estimating the weight of CC, MC, and FL can be seen in Table 14. They increase exponentially from one class to another; in 2020, MC to CC was 4 while FL to CC was 77.

Table 14: Income ratio between different social classes

	2015	2017	2018	2020	2022
MC to CC	3.66	3.91	4.05	4.08	3.96
FL to CC	69.56	74.26	76.97	77.60	93.40

Moreover, the data from the Red Cross did not provide specific details for each municipality in Abidjan, but only for a few of them. The total number of households affected did not match the number of reported household affected for each municipality. For example, in 2018, 375 households were reported to be victims of the floods in Attécoubé and Cocody, but 5000 households were estimated as victims in total. Thus, it was mandatory to find a way to adjust the number of affected households for municipalities other than Attécoubé and Cocody (Table 16). Keeping the example of Year 2018, some of the steps of this adjustment were the following:

1. The clustering based on flood-prone areas, as used in "Method 3: Korea-based" was applied here to estimate the ratio of people affected per municipality. The tiers were defined as follow: TIER 1 = 40%, TIER 2 = 35%, TIER 3 = 15%, TIER 4 = 7%, TIER 5 = 3%
2. These ratios were then matched to the municipalities affected: Abobo was assigned TIER 1 (40%), Adjame was assigned TIER 4 (7%), Attecoubé, Cocody, and Yopougon were assigned TIER 2 (35%), and Koumassi was assigned TIER 3 (15%).
3. The ratio for each cluster was divided by the number of municipalities in that cluster. For example, if TIER 2 includes three municipalities (Attecoubé, Cocody, and Yopougon), the 35% ratio for TIER 2 is divided by 3. The resulting values are shown in the "Ratio per city" column in Table 15.
4. The "Scaling to match" column adapts the ratio per city to include municipalities that were affected but whose damages were not reported. This ensures that the total number of affected households is properly distributed among all municipalities. Thus, the scaling to match value for each municipality represents its share of houses affected. In Year 2018, a total of 5000 households were reported to be affected, but the municipalities of 4625 of them were not specified. Hence, because the scaling to match value for Abobo in 2018 was 0.29, it means that 29% of those 4625 households were in Anyama. It can be checked in Table 16: $\sim 29\% \text{ of } 4625 = \sim 1355 \text{ households}$

Finally, the total amount of construction and demolition debris was estimated to range between 1.7 million and 14 million kilograms.

Table 15: Methodology for adaptation of Red Cross Reports Data

	Ratio per tier		Scaling to match		
			Ratio per city	Year 2018	Year 2020
TIER 1	0.4	Anyama	0.2	N/A	0.27
		Abobo	0.2	0.29	0.27
TIER 2	0.35	Cocody	0.0875	0.13	0.12
		Port Bouet	0.0875	N/A	0.12
		Attecoubé	0.0875	0.13	N/A
		Yopougon	0.0875	0.13	0.12
TIER 3	0.15	Koumassi	0.15	0.22	N/A
		Treichville	0.075	N/A	N/A
TIER 4	0.07	Adjame	0.07	0.10	0.10
TIER 5	0.03	Plateau	0.03	N/A	N/A
<u>TOTAL</u>				1	1

Table 16: Results after adaptation of Red Cross Reports Data

		Reported Households affected	Adjusted Household affected
2018 18th June	Abobo	N/A	1,355
	Adjame	N/A	474
	Attecoubé	155	748
	Cocody	220	813
	Koumassi	N/A	1,016
	Yopougon	N/A	593
2020 15th June	Abobo	600	982
	Adjame	2	136
	Anyama	219	601
	Cocody	N/A	167
	Port Bouet	60	227
	Yopougon	65	232

❖ 2nd Part: Estimation of Consumer Durables (W)

The items that were chosen as typical for CC houses were beds, chairs, couches, tables, small fridges, stoves, basic cutlery, TVs, Radios, and clothes. The respective amount and weight of those typical items for each CC house can be seen in Table 17, while the database for low-end,

mid-end, and high-end furniture created can be seen in Appendix A: The model - Table 38. After applying a +/-20% error margin, the total weight of consumer durables per household was estimated to be 298 kg.

Table 17: Weight and quantity of Consumer Durables

	Item list	number of items per household	average weight (kg)	weight range(kg)	
Furniture (beds, chairs, tables)	bed	2	16	25.1	37.6
	chair + couch	4	16	49.6	74.4
	table	4	18	56	84
Appliances (fridge, stove, other kitchen utensils, TV, radio)	fridge	1	53	42.5	63.8
	stove	1	6	5.1	7.6
	Cutlery (spoon, forks, saucepan, etc)	1	50	40	60
	TV	1	8	6.7	10
	RADIO	1	2	1.6	2.4
	CLOTHES	150	0.1	12.0	18.0
TOTAL				238.5	357.8

Table 18: Total Consumer Durables Waste

		weight range (kg)		CC houses	MC Houses	Total waste (kg)	
Date	Location	min	max			min	max
2018	Abobo	238.5	357.8	100	266	126,920	190,379
18th June	Adjame			35	93	44,422	66,633
	Attecoubé			55	147	70,042	105,064
	Cocody			60	160	76,129	114,194
	Koumassi			75	200	95,190	142,785
	Yopougon			44	117	55,527	83,291
2020	Abobo			117	342	161,197	241,796
15th June	Adjame			16	47	22,275	33,412
	Anyama			72	209	98,654	147,981
	Cocody			20	58	27,433	41,149
	Port Bouet			27	79	37,282	55,923
	Yopougon			28	81	38,103	57,155

2.4. Linear Regression Runs

First Model runs

The linear model run using the parameters from Taiwan was not suitable for the Ivorian dataset. As shown in Table 19, the adjusted R^2 value obtained was 0.185. The exponential one modelled better, with an adjusted R^2 of 0.38. For Korea, the same was done, but with a different number of features. In our case, we tested the same features used in each of those studies on the Abidjan datasets.

After running those models, we then tested different combinations of all the features that were shortlisted in 1.2. Model features selection. We tried combinations of four from the nine features shortlisted on the dataset. This resulted in about $\binom{9}{4} = 126$ model runs. After that, the top five models in terms of adjusted R-squared values were recorded and considered the most efficient four-feature models.

Table 19: First model runs

Runs	Features	Metrics	Source
Run 1	Population Density, Flooded Area, Rainfall	Adj. R^2 = 0.185	Taiwan, linear model [29]
Run 2	Log(Population Density), log(flooded Area), log(rainfall)	Adj. R^2 = 0.38	Taiwan, exponential model [29]
Run 3	Population Density, Flooded Area, Rainfall, Area GDP, Urbanization Ratio, Wind speed	Adj. R^2 = 0.745 (0.592 for exponential)	Korea, linear model [3]
Run 4	Log (Population Density), log (Flooded Area), log(Rainfall), log(Area GDP), log(Urbanization Ratio), log(Wind speed)	Adj. R^2 = 0.681	Korea, exponential model [3]

After running the 126 four-feature models, the five highest adjusted R-squared values were respectively 0.751, 0.733, 0.728, 0.727, and 0.72. As shown in Table 20, the features with the most occurrences at the top were the population density (PD), the urbanization ratio (UR), the wind speed (WIND), and the Total Rainfall (RTOTAL). It should also be noted that a three-feature model ([Models Runs](#)) with PD, WIND, and RTOTAL had a 0.74 adjusted R^2 value.

Table 20: Results of 126 four-feature models runs

Rank	The 4 Features	Adjusted R ²
1	PD + WIND + RTOTAL + UR	0.751
2	PD + UR + ALTMIN + ALTAVG	0.733
3	PD + DURATION + UR + WIND	0.728
4	PD + WIND + RTOTAL + AREA	0.727
5	PD + UR + RTOTAL + DURATION	0.72
...	...	...
122	AREA + GRDP + ALTMIN + ALTMAX	-0.46
123	AREA + GRDP + ALTMIN + PD	-0.461
124	GRDP + PD + ALTMIN + ALTMAX	-0.461
125	AREA + PD + ALTMAX + GRDP	-0.468
126	AREA + PD + ALTMAX + ALTMIN	-0.52

3. Discussion: Further analysis, Interpretation

▪ *Features selection*

The feature selection was significantly affected by the scarcity of relevant data for Cote d'Ivoire. Initially, the plan was to gather as much data as possible for different parameters and use metrics like AIC or BIC to identify the optimal ones. However, it was only possible to obtain or create data for eight parameters. Consequently, these metrics were applied to these eight parameters to study the relevance of each. Obtaining data on other features could therefore provide different results in the future.

The parameters chosen included both pre-disaster and post-disaster features. In the research papers reviewed for the literature review, the focus was solely on either pre-disaster or post-disaster features. In our case, we did not make this separation, and the results we obtained in terms of modelling were not worse than those in other research. Consequently, the difference is surely in the explanation of the results we get. For example, since we used both pre-disaster and post-disaster features, it means that when we evaluate which parameters are more impactful, the procedures we decide to choose to prevent flood waste generation might be trickier than just focusing on pre-disaster or post-disaster prediction.

▪ *Handling insufficient data*

Regarding the handling of insufficient data for the model's features, many estimations were made using contexts different from that of Ivory Coast. Consequently, we cannot be certain about the accuracy of the dataset we created. The government needs to do a better job in providing these datasets, as other countries are already doing well in providing data on these features.

- ***Methods for quantifying the waste***
 - ***Identify key parameters like cc house volumes that can drastically affect the estimation***

The clustering method used to group the municipalities and merge them from one country to another was an effective approach to address the lack of data. However, it is not the optimal method because there are many differences from one country to another, even when they share borders, let alone when they are on different continents.

That is why using Method 5 as the basis and the other ones as a kind of validation to have a range of expected values was the best approach with the available data. As a recall, the goal is not accuracy but to create a framework that can be used to obtain results about flood waste generation. More research published for countries closer to Abidjan will benefit from the framework we created in this study.

Regarding the limitations of using data from Bangkok and Korea, for Bangkok, there are many differences between its locations and Abidjan, and when considering the primary factors influencing flood waste generation, they do not necessarily apply or behave the same in both cities. Nevertheless, there are still many common points between these two cities. Both belong to developing countries with booming economies, have high-density populations, tropical climates, and similar temperatures and rain patterns.

Korea seems to be a more challenging case because we did not focus on just one city but many cities. However, there are still common points between the cities chosen in Korea and the municipalities in Abidjan. They also share similarities in terms of climate.

Identifying one typical CC house is tough because they can vary significantly, and the contents of these houses might differ from one year to another. For that reason, the typical CC house content chosen might differ from reality by an order of magnitude from one year to another. Additionally, 30% and 1% were respectively chosen as the ratio of construction and demolition debris produced after flood waste to the total weight of MC houses and FL houses. Assuming that the weight of a CC house is proportional to the weight of MC houses and FL houses has some limitations, especially given the fact that social classes vary greatly from one municipality to another in Abidjan. Some MC houses are in rich neighbourhoods with poor civil engineering, so estimating that MC houses will not be affected as much as CC houses can be problematic. Still, these cases are very unusual in Abidjan.

Using Jumia for the consumer durables estimation gives a great idea of the mass of the items that turn into waste. Additional data supported by IKEA in terms of item dimensions was also

very useful in estimating the volume of typical house contents. However, it is important to mention that poor people do not buy items on Jumia. Instead, they use second-hand or discarded items from Cote d'Ivoire or waste that is discharged in open dumping sites from all over the world.

Eight features were initially selected for the model, but ultimately, five were deemed sufficient. This reduction from eight to five features is quite good compared to what happened in previous models from other researchers. However, having official data for the chosen parameters could yield different results regarding which features are more important. For example, actual urbanization ratios and GDP per area values for Abidjan might show less correlation and thus both be needed in the model.

Regarding the summary of the quantification results, the summary statistics show that the spread for Abidjan and Bangkok are very different. This is explained by the fact that, even though these cities have many common points, they are very far apart in terms of geographical location, population type, and population habits. In the future, obtaining data for Abidjan and Bangkok from more relevant countries or cities will be better. The method chosen to decide which municipalities have higher exposure seems realistic compared to other research papers, but more accuracy is needed for greater relevance.

As we can see in the summary tables, the differences in terms of weight generation are closer for Method 1 and Method 2 together, Method 3 and Method 4 together, while Method 5 is far away from the other four methods. As mentioned earlier, the basic prediction is from Method 5, and the spread is estimated with the other four methods.

Regarding the point of identifying what happened in the municipalities of Korea in 2012 to what happened in Abidjan in 2015, it is not the most appropriate comparison, yet it is still useful to update what happened somewhere and use it to simulate for other countries. Since the goal of this research is to create a framework, even using data from 1999, the most important aspect was to develop a prototype of the study method we used to get results. Many research studies have been conducted using similar methods. Assuming a steady and consistent percentage increase in waste generation from 1989 to 2016 can seem problematic. Indeed, waste generation depends on many factors that have certainly changed over that 27-year span, yet it was still a good approximation of the waste increase given the lack of data.

The backup results are similar despite using different approaches. This is mainly due to the fact that the key parameters influencing the final flood waste generation we used in the study were quite close. With different assumptions, we could have obtained a higher range of results. This is the same for Korea. However, the number of days was not included in the calculation for flood waste using Korean data because it was not mentioned in the research paper about floods in Korea. Therefore, the results might have been closer to what we observed in Bangkok or Abidjan if we had more data on the duration of the floods.

Finally, the results we obtained by changing some of the parameters could affect the final flood waste prediction, especially when we look at the assumed values for certain parameters like

the volume of one CC house, the materials used as main components with different densities depending on the material chosen, and the fact that other buildings like schools, clinics, local shops, street food stalls, and items placed on the roads were not taken into account as flood wastes generated.

Table 21: Total Flood waste composition

		Range: From 2018 to 2021	
Category	Materials	Proportion from	Proportion from
Construction and Demolition Debris	cement	8	98.65 - 99.15
	Mud	77	
	Aluminium	14	
Consumer Durables	Wool &	34	0.85 – 1.35
	Soft wood	32	
	Stainless steel	20	
	Plastic casing	6	
	cotton	7	
Consumer durables:	Copper, Lead,	-	1.2 – 1.9

Table 22: Comparative summary for all 5 methods

	<u>Method 1</u>	<u>Method 2</u>	<u>Method 3</u>	<u>Method 4</u>	<u>Method 5</u>
Min	70,000	40,000	2,000,000	7,000,000	1,700,000
Max	7,600,000	6,960,000	69,200,000	33,000,000	22,400,000
Median	610,000	400,000	5,600,000	20,000,000	3,500,000
Average	1,500,000	1,000,000	19,600,000	20,200,000	6,150,000
Standard deviation	2,100,000	1,700,000	22,700,000	6,400,000	4,700,000

4. Conclusions and Limitations

The methodology followed enabled the development of a quantitative prediction model for flood waste generation in Abidjan, based on the Population Density, Flooded Area, Rainfall, Area GDP, Urbanization Ratio, Wind speed. The quantification of the waste using five different methods allowed us to estimate a range for the flood waste generated in Abidjan. Bayesian inference provided additional insights into the probabilities involved for each parameters based on the deterministic regression results. This study has many limitations that make the numbers obtained not necessarily trustworthy. The validation was difficult to conduct because of the lack of similar research work. Thus, the features selected might not be the most appropriate and certain underlying factors might have not been captured. Nevertheless, the framework used will allow to find significantly more accurate results once real data is available.

IV. Environmental Impact Assessment

1. Methodology

The environmental impact assessment in the context of this study focuses on: (1) Estimating the space required for a temporary disposal of the flood waste; (2) Addressing the discharge of heavy metals contained in the flood waste

1.1. Space required

The space required for temporary disposing the flood waste depends on many factors [5]. The design of the site, especially for large disasters, dictates the efficiency and the success of the management site operations. An extensive accumulation of debris should be avoided at temporary sites due to environmental and safety concerns mainly. In Abidjan, even though the flood events are not large scale disasters, there are very few landfill sites in Abidjan. Thus, it is relevant to consider temporary sites if the issue of waste wants to be taken seriously.

In the case of this study, the focus for the space required for the temporary site will be put on the volume of waste. From the first part ([The model](#)), we found the flood waste composition to be consisted of construction and demolition debris, and consumer durables. Since that estimation was in terms of mass, the flood waste density will be used to derive the total flood waste volume from the estimated mass. Regarding the consumer durables taken from the Jumia website mentioned earlier in the first part, many of them had information about their dimensions. In those cases, finding the density is not necessary, as the volume can be directly derived from the size.

1.2. Heavy metals

We could not find a clear consensus on what a heavy metal is as it depends on the context [30]. Most of the time, they are categorized as metals with high density (5g/cm³) [31] or metals categorized as dangerous for human health. In the context of this thesis, the focus was put on the ones considered as dangerous or problematic. To do so, the list of heavy metals considered was decided based on:

- Basel Convention regulations

Cote d'Ivoire, which capital city is Abidjan, has been linked to the Basel convention since 1994, December 1st. According to their guidelines [32] [33], the presence of a (heavy) metal does not necessarily mean that it is in its hazardous form. For example, scrap metallic lead in non-dispersible form is not considered hazardous. From their List A1010 in ANNEX VIII, should be considered hazardous some metal wastes and waste consisting of alloys of certain components. However, some of them can be excluded from the list if they fit the criteria of items listed in the LIST B from ANNEX VIII.

- Research articles about heavy metals usually found in the potable source of water in Abidjan and the Ebrie Lagoon [12],[34].

2. Results and Discussion

2.1. Space required

The corresponding densities to the construction materials and consumer durables are registered, respectively, in Table 23 and

Table 24. As a recall, one main assumption is that all the slum houses (CC houses) are destroyed and turn out to debris.

The volume of the construction and demolition debris for one CC house was estimated to be on average 8.2 m³. The typical materials used to build one house in a slum were assumed to be Mud/Earth for the building materials, Portland Cement for the binding materials, Aluminium sheets for the roof, and water as main component for liquid & additives. The estimation of the mass of the construction items can be seen in Appendix B : EIA. A 20% margin was applied to the actual volume calculated, resulting in “volume min” and “volume max” values (Table 23). The typical consumer durables with their mass and dimensions can be seen in Table 24.

Table 23: Volume of Construction and Demolition Debris for CC houses

ITEMS	DENSITY (kg/m3)	MASS min (kg)	MASS max(kg)	VOLUME min (m3)	VOLUME max (m3)
Building material	1890	9677	14515	5.12	7.68
Binding material	1506	964	1446	0.64	0.96
Roof	2739	1753	2629	0.64	0.96
liquids/additives	1000	124	186	0.12	0.185904

Table 24: Typical consumer durables for CC houses

PLAIN ITEMS	Unit number	DENSITY (kg/m3)	
bed	4	611	wool, polyurethane
couch	1	54	triple size ikea sofa divided by 3
chair	3	235	ikea chair dimensions for volume
table	4	326	ikea
fridge	1	1557	40% Al / 30% polyurethane foam for insulation / 20% plastic for lining / 10% for refrigerator components - according to chat GPT
stove	1	7850	steel for 90% of the body
cutlery	1	7850	stainless steel
TV	1	1150	40% plastic casing // 40% aluminium metal framework // glass screen 15% // others 5%
Radio	1	5000	unknown
bedding + Clothes (L size)	250	24	50% polyester/ 120 g /m2 and 2 clothes per 1cm
screws for wood tables	50	7850	stainless steel

Hence, the total waste volume that will be stored in multiple temporary locations after floods Abidjan municipalities could range between 2032m³ and 4488m³.

		Houses destroyed	min Total Volume (m3)	max Total Volume (m3)
2018 18th June	Abobo	100	811	1217
	Adjame	35	284	426
	Attecoubé	55	448	671
	Cocody	60	486	730
	Koumassi	75	608	912
	Yopougon	44	355	532
2020 15th June	Abobo	117	944	1417
	Adjame	16	131	196
	Anyama	72	578	867
	Cocody	20	161	241
	Port Bouet	27	218	328
	Yopougon	28	223	335

2.2. Heavy metals

The complete list of 15 heavy metals from the different relevant sources to consider was: Antimony, Arsenic, Beryllium, Cadmium, Lead, Mercury, Selenium, Tellurium, Thallium, Zinc, Copper, Chromium, Mercury, Nickel and Manganese.

From that list, only 6 were part of the flood waste in the case of Abidjan according to the waste estimation: Copper, Lead, Steel, Mercury, Nickel, Zinc and Aluminium. The provenance of each of them can be seen in Table 25 below. The other heavy metals not selected are either non-existent, or in extremely small proportion.

Note: Because of the lack of data, some adjustments were made to estimate the amount of heavy metals in fridges and TVs. They were derived from a survey study on average metal content from various e-waste [35] .

Table 25: The heavy metals and their provenance

ITEMS	Heavy Metals						
	Copper	Lead	steel	Mercury	Nickel	Zinc	Aluminium
Roof							x
Liquids /additives					x		
fridge	x	x			x	x	
stove			x				

cutlery			x				
TV	x				x	x	x
Radio	x						
electrical equipment				x			
screws			x				

The metal with the highest amount was Aluminium, corresponding to more than 95% of all metals present. It comes from the fact that it is the main material for roofs, which are a huge part of the C&D debris. Steel and Zinc also had noticeable proportions and mainly come from cutlery, screws, and fridges.

The quantities of all the metals in ppm (parts per million) are shown in Table 26. According to regulations, the values obtained are not critical. Critical components like Lead and Mercury have respectively 2 and 0.0021 ppm, which are far below the environmental & Landfill guidelines limits registered in [36] *Basel Convention Technical Guidelines* (2004).

Table 26: Percentage of Metals from Total Waste

METALS	Mass (kg) 2018	ppm	Mass (kg) 2020	ppm
Copper	11724	212	8838	310
Lead	115	2	86	3
Steel	16507	299	12443	436
Mercury	0.12	0.0021	0.09	0.0031
Nickel	2863	52	2158	76
Zinc	2297	42	1732	61
Aluminium	689199	12,487	519530	18,203
TOTAL METALS	722,705	13,094	544,788	19,088
TOTAL WASTE	55,191,687	1,000,000	28,540,321	1,000,000

ppm = Parts Per Million

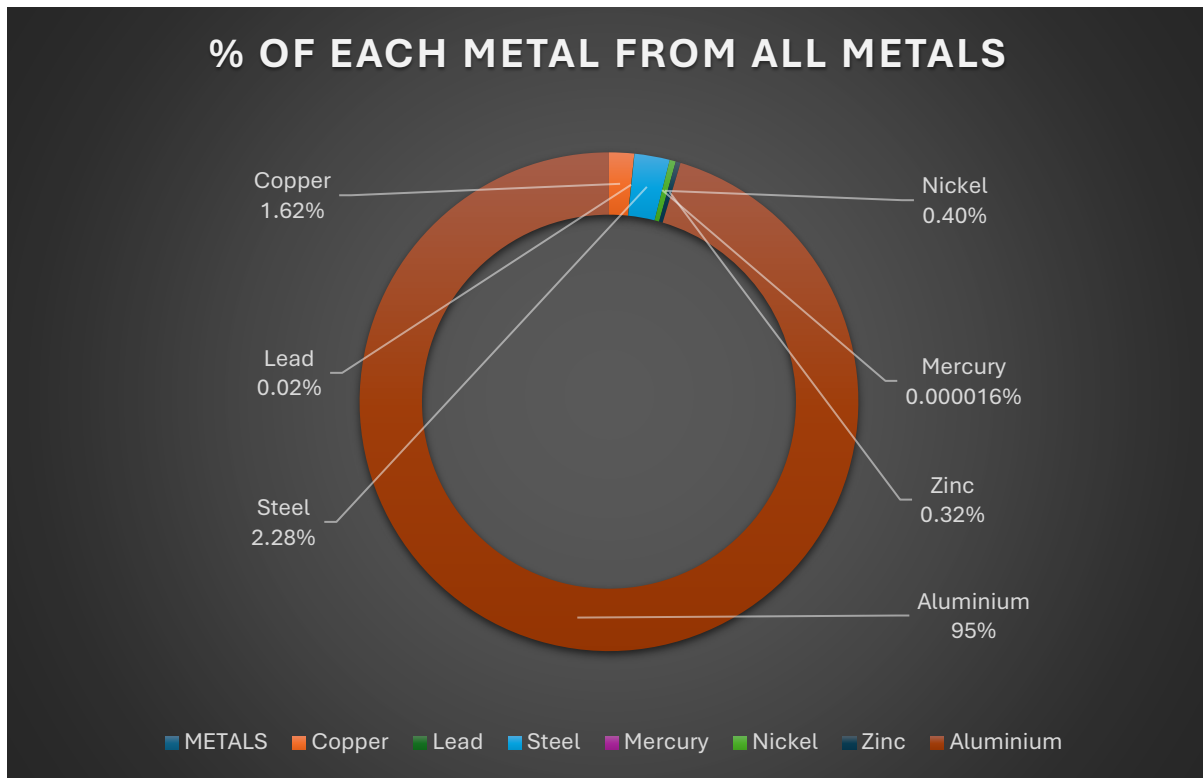


Figure 9: Heavy metals in flood waste

3. Conclusions and Limitations

A framework was developed to estimate the total waste volume after each flood event and the heavy metals contained in it. Additional heavy metals that could be seen in Abidjan that were captured in the flood waste estimation model were also taken into account. Considering that this second section is closely related to the results of the first one, better data for the first part will make the EIA estimations more accurate. Nevertheless, even though the quantity of sensible heavy metals like Lead and Mercury was overestimated, their amounts found were still in small proportion relatively to the environmental restriction limits usually observed internationally. Finally, it is good to recall that there are more considerations to take into account when we consider the space for temporary disposal than the volume of the waste only.

V. Conclusion

In the first part of this work, the methodology followed enabled the development of a quantitative prediction model for flood waste generation in Abidjan, based on the Population Density, Flooded Area, Rainfall, Area GDP, Urbanization Ratio, and Wind speed. The quantification of the waste using five different methods allowed us to estimate a range for the flood waste generated in Abidjan. Bayesian inference provided additional insights into the probabilities involved for each parameters based on the deterministic regression results. This study has many limitations that make the numbers obtained not necessarily trustworthy. The

validation was difficult to conduct because of the lack of similar research work. Thus, the features selected might not be the most appropriate and certain underlying factors might have not been captured. Nevertheless, the framework used will allow to find significantly more accurate results once real data is available.

In the second part, the framework developed permitted to estimate the heavy metals contained in the flood waste. Additional heavy metals that could be present in Abidjan that were captured in the flood waste estimation model were also considered. Although the quantity of sensible heavy metals like Lead and Mercury was overestimated, their calculated amounts were still in small proportion relatively to the usual environmental restriction limits. The two sections are closely related, so better data for the first part will make the EIA estimations more accurate.

For future work, it will be crucial to have much more access to real data from Abidjan. This could be achieved through surveys with local/government authorities and fieldwork. The existing data could also be enhanced by new data augmentation algorithms provided by the advancements in Artificial Intelligence [37]. These extended datasets would allow the use of advanced machine learning algorithms like neural networks for the prediction of flood waste in Abidjan. More importantly, incentives must be created in Cote d'Ivoire to generate more interest from the research community on this matter.

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VII. Appendices

1. Appendix A: The model

Table 27: Database Thailand

District	(a) Average waste amount during normal situation (TONS)	(b) Total waste amount from 2011/12/1 to 2012/1/1	Flood waste (=b)-(a)*60) (tons)	Flood waste per house-hold (kg)	Household number
Don Mueang	157	29642.74	20222.74	313	64609.39297
Lak Si	147	13268.14	4448.14	96.7	45999.37952
Sai Mai	166	18037.13	8077.13	94.5	85472.27513
Bang Khen	237	17955.3	3735.3	41.6	89790.86538
Chatuchak	338	23609.39	3229.39	25.9	124686.8726
Min Buri	169	10915.25	775.25	15.4	50340.90909
Khlong Sam Wa	127	8567.5	947.5	14.9	63590.60403
Nong Chok	88	5682.04	402.04	7.9	50891.13924
Khan Na Yao	98	7724.28	1844.28	53.4	34537.07865

Bang Khae	258	19622.16	4142.16	53.5	77423.5514
Bang Phlat	143	10532.28	1952.28	43.2	45191.66667
Taling Chan	119	7677.55	897.55	24.2	37088.84298
Phasi Charoen	149	10368.99	1448.99	31.3	46293.61022
Nong Khaem	164	12136.88	2296.88	41.9	54818.13842
Thawi Watthana	92	6579.84	1059.84	35.5	29854.64789
Bangkok Noi	186	11857.77	697.77	15.7	44443.94904
Bangkok Yai	88	5410.72	130.72	4.9	26677.55102
Bang Bon	187	11267.78	47.78	1.1	43436.36364
Lat Krabang	203	12615.37	435.37	6.1	71372.13115
Lat Phrao	157	9781.29	361.29	7.2	50179.16667
Chom Thong	192	11024.47	N/A	N/A	N/A
Bang Khun Thian	256	14957.11	N/A	N/A	N/A

Bang Kapi	284	18690.91	1650.91	18.7	88283.95722
Bueng Kum	151	10434.29	1374.29	21.8	63040.82569
Thon Buri	163	9558.52	N/A	N/A	N/A
Khlong Toei	283.64	18324.08	1316.54	23.1	56993.07359
Khlong San	121.18	7386.14	115.34	3.4	33923.52941
Din Daeng	230.53	15149.36	1317.56	25.4	51872.44094
Dusit	176.09	10670.05	104.65	3.3	31712.12121
Thung Khru	115.54	6279.52	N/A	N/A	N/A
Bang Kho Laem	125.32	7109.97	N/A	N/A	N/A
Bang Sue	158.28	9818.41	321.61	7.1	45297.1831
Bang Na	189.84	11600.77	497.37	5.1	97523.52941
Bang Rak	152.56	9794.88	641.28	24.9	25754.21687
Pathum Wan	218.5	15653.47	2543.47	97.6	26060.14344

Prawet	214.64	12947.04	68.64	1	68640
Pom Prap Sattru Phai	87.24	5365.83	131.43	6.8	19327.94118
Phaya Thai	148.87	9853.58	921.38	25.8	35712.4031
Phra Nakhon	135.57	11151.34	3017.14	163.4	18464.74908
Phra Khanong	190.53	8661.33	N/A	N/A	N/A
Yan Nawa	159.12	9869.73	322.53	7.5	43004
Ratchathewi	173.99	11625.57	1186.17	30.8	38512.01299
Rat Burana	118.2	6704.02	N/A	N/A	N/A
Wang Thonglang	195.12	12968.18	1260.98	22.8	55306.14035
Watthana	232.3	15171.61	1233.61	22.1	55819.45701
Saphan Sung	187.19	6071.93	N/A	N/A	N/A
Sathon	155.27	10357.28	1041.08	29.2	35653.42466
Samphanthawong	58.2	3757.87	265.87	20	13293.5

Suan Luang	187.19	12347.63	1116.23	20.9	53408.13397
Huai Khwang	149.41	10615.56	1650.96	32.8	50334.14634
Other public sectors		73657.62	73657.62	N/A	N/A

1.1.Method 1

Table 28: Flood time vs normal time – Thailand

District	(a) Average waste amount during normal situation (TONS)	(b) Total waste amount from 2011/12/1 to 2012/1/1	Flood waste (=b)-(a)*60) (tons)	RATIO ABNORMAL TO NORMAL
<i>Don Mueang</i>	157	29642.74	20222.74	3.15
<i>Sai Mai</i>	166	18037.13	8077.13	1.81
<i>Lak Si</i>	147	13268.14	4448.14	1.50
<i>Phra Nakhon</i>	135.57	11151.34	3017.14	1.37
<i>Khan Na Yao</i>	98	7724.28	1844.28	1.31
<i>Bang Khae</i>	258	19622.16	4142.16	1.27
<i>Bang Khen</i>	237	17955.3	3735.3	1.26
<i>Nong Khaem</i>	164	12136.88	2296.88	1.23
<i>Bang Phlat</i>	143	10532.28	1952.28	1.23

<i>Pathum Wan</i>	218.5	15653.47	2543.47	1.19
<i>Thawi Watthana</i>	92	6579.84	1059.84	1.19
<i>Huai Khwang</i>	149.41	10615.56	1650.96	1.18
<i>Phasi Charoen</i>	149	10368.99	1448.99	1.16
<i>Chatuchak</i>	338	23609.39	3229.39	1.16
<i>Bueng Kum</i>	151	10434.29	1374.29	1.15
<i>Taling Chan</i>	119	7677.55	897.55	1.13
<i>Khlong Sam Wa</i>	127	8567.5	947.5	1.12
<i>Ratchathewi</i>	173.99	11625.57	1186.17	1.11
<i>Sathon</i>	155.27	10357.28	1041.08	1.11
<i>Wang Thonglang</i>	195.12	12968.18	1260.98	1.11
<i>Phaya Thai</i>	148.87	9853.58	921.38	1.10
<i>Suan Luang</i>	187.19	12347.63	1116.23	1.10

<i>Bang Kapi</i>	284	18690.91	1650.91	1.10
<i>Din Daeng</i>	230.53	15149.36	1317.56	1.10
<i>Watthana</i>	232.3	15171.61	1233.61	1.09
<i>Khlong Toei</i>	283.64	18324.08	1316.54	1.08
<i>Min Buri</i>	169	10915.25	775.25	1.08
<i>Nong Chok</i>	88	5682.04	402.04	1.08
<i>Samphanthawong</i>	58.2	3757.87	265.87	1.08
<i>Bang Rak</i>	152.56	9794.88	641.28	1.07
<i>Bangkok Noi</i>	186	11857.77	697.77	1.06
<i>Bang Na</i>	189.84	11600.77	497.37	1.04
<i>Lat Phrao</i>	157	9781.29	361.29	1.04
<i>Lat Krabang</i>	203	12615.37	435.37	1.04
<i>Bang Sue</i>	158.28	9818.41	321.61	1.03

<i>Yan Nawa</i>	159.12	9869.73	322.53	1.03
<i>Pom Prap Sattru Phai</i>	87.24	5365.83	131.43	1.03
<i>Bangkok Yai</i>	88	5410.72	130.72	1.02
<i>Khlong San</i>	121.18	7386.14	115.34	1.02
<i>Dusit</i>	176.09	10670.05	104.65	1.01
<i>Prawet</i>	214.64	12947.04	68.64	1.01
<i>Bang Bon</i>	187	11267.78	47.78	1.00

Table 29: Database KOREA

Year	Special city or province	Municipality	Flood waste (tons)	Population (2015)	year	area (km2)	density	PER PERSON WASTE
2012	Jellabuk-do	Gunsan-si	2,365	278505	2015	395.9	703.4730993	0.00849
2012	Jellabuk-do	Iksan-si	182	307000	2016	507.1	605.4032735	0.00059
2012	Jellabuk-do	Jeongeup-si	220	110194	2015	692.7	159.0789664	0.00200
2012	Jellabuk-do	Namwon-si	168	81257	2020	752.6	107.9683763	0.00207
2012	Jellabuk-do	Gimje-si	400	88479	2016	545.2	162.287234	0.00452
2012	Jellabuk-do	Wanju-gun	430	92220	2020	820.9	112.3401145	0.00466
2012	Jellanam-do	Mokpo-si	530	224509	2020	51.64	4347.579396	0.00236
2012	Jellanam-do	Suncheon-si	150	279435	2016	907.4	307.9512894	0.00054
2012	Jellanam-do	Gokseong-gun	240	31009	2014	547.4	56.64778955	0.00774
2012	Jellanam-do	Gurye-gun	600	27115	2014	443.02	61.20491174	0.02213
2012	Jellanam-do	Goheung-gun	1,050	70000	2015	807.2	86.71952428	0.01500

2012	Jellanam-do	Boseong-gun	116	37888	2020	663.4	57.11184806	0.00306
2012	Jellanam-do	Hwasun-gun	379	66900	2014	786.9	85.01715593	0.00567
2012	Jellanam-do	Jangheung-gun	1,182	35273	2020	622.4	56.67255784	0.03351
2012	Jellanam-do	Gangjin-gun	1,900	32778	2020	501.8	65.32084496	0.05797
2012	Jellanam-do	Yeongam-gun	1,000	57076	2020	611.6	93.32243296	0.01752
2012	Jellanam-do	Muan-gun	1,366	84637	2020	456.9	185.2418472	0.01614
2012	Jellanam-do	Yeonggwang-gun	180	55016	2018	473.7	116.1410175	0.00327
2012	Jellanam-do	Jangseong-gun	574	41443	2020	518.6	79.91322792	0.01385
2012	Jellanam-do	Wando-gun	601	48269	2020	391.8	123.1980602	0.01245
2012	Jellanam-do	Jindo-gun	430	29297	2020	440.1	66.5689616	0.01468
2012	Jellanam-do	Sinan-gun	2,170	35138	2020	678.9	51.75725438	0.06176
2012	Ulsan	Dong-gu	80	232559	2020	137.2	1695.036443	0.00034
2012	Ulsan	Ulju-gun	428	225050	2020	756.5	297.4884336	0.00190

2012	Jellanam-do	Naju-si	80	113293	2020	605.8	187.013866	0.00071
2012	Jellanam-do	Haenam-gun	2,700	65223	2020	1048	62.23568702	0.04140
2012	Jellanam-do	Hampyeong-gun	100	39156	2020	77.34	506.283941	0.00255
2012	Jellanam-do	Jangseong-gun	274	41443	2020	518	80.00579151	0.00661
2012	Gyeongsangbuk-do	Pohang-si	3,703	501109	2020	442.2	1133.218001	0.00739
2012	Gyeongsangbuk-do	Gimcheon-si	3,500	139145	2020	138.4	1005.382948	0.02515
2012	Gyeongsangbuk-do	Goryeong-gun	2,000	31419	2020	386.7	81.24903026	0.06366
2012	Gyeongsangbuk-do	Seongju-gun	3,500	13533	2020	36.18	374.0464345	0.25863
2012	Gyeongsangnam-do	Miryang-si	1,100	103228	2020	801.4	128.8095832	0.01066
2012	Gyeongsangnam-do	Yongsan-si	1,600	351206	2020	721.7	486.6371068	0.00456
2012	Gyeongsangnam-do	Hadong-gun	230	41772	2020	678.8	61.53800825	0.00551
2012	Gyeongsangnam-do	Sancheong-gun	250	33500	2020	792	42.2979798	0.00746
2012	Gyeongsangnam-do	Hapcheon-gun	173	42140	2020	983.8	42.83390933	0.00411

Table 30: Method 1 - Clustering of ratios

District	Flood waste per household (kg)	Ratio normal/total	Clusters	Average
Don Mueang	313	3.15	TIER 3	1.83
Sai Mai	94.5	1.81		
Lak Si	96.7	1.50		
Phra Nakhon	163.4	1.37		
Khan Na Yao	53.4	1.31		
Bang Khae	53.5	1.27	TIER 2	1.25
Bang Khen	41.6	1.26		
Nong Khaem	41.9	1.23		
Bang Phlat	43.2	1.23		
Pathum Wan	97.6	1.19	TIER 1	1.09

Thawi Watthana	35.5	1.19		
Huai Khwang	32.8	1.18		
Phasi Charoen	31.3	1.16		
Chatuchak	25.9	1.16		
Bueng Kum	21.8	1.15		
Taling Chan	24.2	1.13		
Khlong Sam Wa	14.9	1.12		
Ratchathewi	30.8	1.11		
Sathon	29.2	1.11		
Wang Thonglang	22.8	1.11		
Phaya Thai	25.8	1.10		
Suan Luang	20.9	1.10		
Bang Kapi	18.7	1.10		

Din Daeng	25.4	1.10		
Watthana	22.1	1.09		
Khlong Toei	23.1	1.08		
Min Buri	15.4	1.08		
Nong Chok	7.9	1.08		
Samphanthawong	20	1.08		
Bang Rak	24.9	1.07		
Bangkok Noi	15.7	1.06		
Bang Na	5.1	1.04		
Lat Phrao	7.2	1.04		
Lat Krabang	6.1	1.04		
Bang Sue	7.1	1.03		
Yan Nawa	7.5	1.03		

Pom Prap Sattru Phai	6.8	1.03		
Bangkok Yai	4.9	1.02		
Khlong San	3.4	1.02		
Dusit	3.3	1.01		
Prawet	1	1.01		
Bang Bon	1.1	1.00		

District	Flood waste per household (kg)	Ratio normal/total	Clusters	Average
Don Mueang	313	3.15	TIER 3	1.83
Sai Mai	94.5	1.81		
Lak Si	96.7	1.50		
Phra Nakhon	163.4	1.37		
Khan Na Yao	53.4	1.31		
Bang Khae	53.5	1.27	TIER 2	1.25
Bang Khen	41.6	1.26		
Nong Khaem	41.9	1.23		
Bang Phlat	43.2	1.23		
Pathum Wan	97.6	1.19	TIER 1	1.09
Thawi Watthana	35.5	1.19		
Huai Khwang	32.8	1.18		
Phasi Charoen	31.3	1.16		
Chatuchak	25.9	1.16		
Bueng Kum	21.8	1.15		
Taling Chan	24.2	1.13		
Khlong Sam Wa	14.9	1.12		
Ratchathewi	30.8	1.11		
Sathon	29.2	1.11		
Wang Thonglang	22.8	1.11		
Phaya Thai	25.8	1.10		
Suan Luang	20.9	1.10		
Bang Kapi	18.7	1.10		
Din Daeng	25.4	1.10		
Watthana	22.1	1.09		

Khlong Toei	23.1	1.08		
Min Buri	15.4	1.08		
Nong Chok	7.9	1.08		
Samphanthawong	20	1.08		
Bang Rak	24.9	1.07		
Bangkok Noi	15.7	1.06		
Bang Na	5.1	1.04		
Lat Phrao	7.2	1.04		
Lat Krabang	6.1	1.04		
Bang Sue	7.1	1.03		
Yan Nawa	7.5	1.03		
Pom Prap Sattru Phai	6.8	1.03		
Bangkok Yai	4.9	1.02		
Khlong San	3.4	1.02		
Dusit	3.3	1.01		
Prawet	1	1.01		
Bang Bon	1.1	1.00		

Table 31: Method 2 - Average waste per household

	Flood Waste per household Thailand (kg)	Tiers	Tier Averages Flood Waste per household Thailand (kg)
Prawet	1.0	5	5.12
Bang Bon	1.1		

Dusit	3.3		
Khlong San	3.4		
Bangkok Yai	4.9		
Bang Na	5.1		
Lat Krabang	6.1		
Pom Prap Sattru Phai	6.8		
Bang Sue	7.1		
Lat Phrao	7.2		
Yan Nawa	7.5		
Nong Chok	7.9		
Khlong Sam Wa	14.9	4	16.94
Min Buri	15.4		
Bangkok Noi	15.7		
Bang Kapi	18.7		
Samphanthawong	20.0		
Suan Luang	20.9	3	26.43
Bueng Kum	21.8		
Wattana	22.1		
Wang Thonglang	22.8		
Khlong Toei	23.1		
Taling Chan	24.2		
Bang Rak	24.9		
Din Daeng	25.4		
Phaya Thai	25.8		
Chatuchak	25.9		
Sathon	29.2		
Ratchathewi	30.8		
Phasi Charoen	31.3		

Huai Khwang	32.8	2	46.72
Thawi Watthana	35.5		
Bang Khen	41.6		
Nong Khaem	41.9		
Bang Phlat	43.2		
Khan Na Yao	53.4		
Bang Khae	53.5		
Sai Mai	94.5	1	153.04
Lak Si	96.7		
Pathum Wan	97.6		
Phra Nakhon	163.4		
Don Mueang	313.0		

Table 32: Method 3

Year	Special city or province	Municipality	Flood waste (tons)	Population (2015)	year	Waste (tons) /person
2012	Jellabuk-do	Gunsan-si	2,365	278505	2015	0.00849
2012	Jellabuk-do	Iksan-si	182	307000	2016	0.00059

2012	Jellabuk-do	Jeongeup-si	220	110194	2015	0.00200
2012	Jellabuk-do	Namwon-si	168	81257	2020	0.00207
2012	Jellabuk-do	Gimje-si	400	88479	2016	0.00452
2012	Jellabuk-do	Wanju-gun	430	92220	2020	0.00466
2012	Jellanam-do	Mokpo-si	530	224509	2020	0.00236
2012	Jellanam-do	Suncheon-si	150	279435	2016	0.00054
2012	Jellanam-do	Gokseong-gun	240	31009	2014	0.00774
2012	Jellanam-do	Gurye-gun	600	27115	2014	0.02213
2012	Jellanam-do	Goheung-gun	1,050	70000	2015	0.01500
2012	Jellanam-do	Boseong-gun	116	37888	2020	0.00306
2012	Jellanam-do	Hwasun-gun	379	66900	2014	0.00567
2012	Jellanam-do	Jangheung-gun	1,182	35273	2020	0.03351
2012	Jellanam-do	Gangjin-gun	1,900	32778	2020	0.05797
2012	Jellanam-do	Yeongam-gun	1,000	57076	2020	0.01752
2012	Jellanam-do	Muan-gun	1,366	84637	2020	0.01614
2012	Jellanam-do	Yeonggwang-gun	180	55016	2018	0.00327
2012	Jellanam-do	Jangseong-gun	574	41443	2020	0.01385
2012	Jellanam-do	Wando-gun	601	48269	2020	0.01245
2012	Jellanam-do	Jindo-gun	430	29297	2020	0.01468
2012	Jellanam-do	Sinan-gun	2,170	35138	2020	0.06176
2012	Ulsan	Dong-gu	80	232559	2020	0.00034
2012	Ulsan	Ulju-gun	428	225050	2020	0.00190
2012	Jellanam-do	Naju-si	80	113293	2020	0.00071
2012	Jellanam-do	Haenam-gun	2,700	65223	2020	0.04140
2012	Jellanam-do	Hampyeong-gun	100	39156	2020	0.00255
2012	Jellanam-do	Jangseong-gun	274	41443	2020	0.00661
2012	Gyeongsangbuk-do	Pohang-si	3,703	501109	2020	0.00739
2012	Gyeongsangbuk-do	Gimcheon-si	3,500	139145	2020	0.02515
2012	Gyeongsangbuk-do	Goryeong-gun	2,000	31419	2020	0.06366

2012	Gyeongsangbuk-do	Seongju-gun	3,500	13533	2020	0.25863
2012	Gyeongsangnam-do	Miryang-si	1,100	103228	2020	0.01066
2012	Gyeongsangnam-do	Yangsan-si	1,600	351206	2020	0.00456
2012	Gyeongsangnam-do	Hadong-gun	230	41772	2020	0.00551
2012	Gyeongsangnam-do	Sancheong-gun	250	33500	2020	0.00746
2012	Gyeongsangnam-do	Hapcheon-gun	173	42140	2020	0.00411

1.2.Method 2

	Flood Waste per household Thailand (kg)	Tiers	Tier Averages Flood Waste per household Thailand (kg)
Prawet	1.0	5	5.12
Bang Bon	1.1		

Dusit	3.3		
Khlong San	3.4		
Bangkok Yai	4.9		
Bang Na	5.1		
Lat Krabang	6.1		
Pom Prap Sattru Phai	6.8		
Bang Sue	7.1		
Lat Phrao	7.2		
Yan Nawa	7.5		
Nong Chok	7.9		
Khlong Sam Wa	14.9	4	16.94
Min Buri	15.4		
Bangkok Noi	15.7		
Bang Kapi	18.7		
Samphanthawong	20.0		
Suan Luang	20.9	3	26.43
Bueng Kum	21.8		
Watthana	22.1		
Wang Thonglang	22.8		
Khlong Toei	23.1		
Taling Chan	24.2		
Bang Rak	24.9		
Din Daeng	25.4		
Phaya Thai	25.8		
Chatuchak	25.9		
Sathon	29.2		
Ratchathewi	30.8		

Phasi Charoen	31.3		
Huai Khwang	32.8		
Thawi Watthana	35.5		
Bang Khen	41.6	2	46.72
Nong Khaem	41.9		
Bang Phlat	43.2		
Khan Na Yao	53.4		
Bang Khae	53.5		
Sai Mai	94.5	1	153.04
Lak Si	96.7		
Pathum Wan	97.6		
Phra Nakhon	163.4		
Don Mueang	313.0		

1.3.Method 3

Table 33:

Year	Location	Population	Flood Waste (kg)
2015	Adjame	285,492	4,681,312

	Attecoubé	262,246	4,300,138
2017	Abobo	1,190,573	62,241,510
	Koumassi	366,307	2,480,520
	Yopougon	1,395,871	23,164,094
2018	Abobo	1,226,290	64,493,408
	Adjame	311,964	5,208,027
	Attecoubé	286,563	4,783,965
	Cocody	633,812	1,961,404
	Koumassi	377,296	2,570,265
	Yopougon	1,437,747	24,002,171
2020	Abobo	1,300,971	69,244,572
	Adjame	330,963	5,591,697
	Anyama	378,245	20,132,188
	Cocody	672,411	2,105,899
	Port Bouet	600,772	10,150,176
	Yopougon	1,525,306	25,770,387

Table 34: Clustering KOREA

	Municipality	Waste (tons) /person	Waste (kg) /person	Average Waste (tons)/person	TIERS
1	Dong-gu	0.00034	0.34	0.000545	TIER 5
2	Suncheon-si	0.00054	0.54		
3	Iksan-si	0.00059	0.59		
4	Naju-si	0.00071	0.71		
5	Ulju-gun	0.00190	1.90	0.003040	TIER 4
6	Jeongeup-si	0.00200	2.00		
7	Namwon-si	0.00207	2.07		
8	Mokpo-si	0.00236	2.36		
9	Hampyeong-gun	0.00255	2.55		
10	Boseong-gun	0.00306	3.06		
11	Yeonggwang-gun	0.00327	3.27		
12	Hapcheon-gun	0.00411	4.11		
13	Gimje-si	0.00452	4.52		
14	Yangsan-si	0.00456	4.56		
15	Wanju-gun	0.00466	4.66	0.006691	TIER 3
16	Hadong-gun	0.00551	5.51		
17	Hwasun-gun	0.00567	5.67		
18	Jangseong-gun	0.00661	6.61		
19	Pohang-si	0.00739	7.39		
20	Sancheong-gun	0.00746	7.46		
21	Gokseong-gun	0.00774	7.74		
22	Gunsan-si	0.00849	8.49		

23	Miryang-si	0.01066	10.66	0.016397	TIER 2
24	Wando-gun	0.01245	12.45		
25	Jangseong-gun	0.01385	13.85		
26	Jindo-gun	0.01468	14.68		
27	Goheung-gun	0.01500	15.00		
28	Muan-gun	0.01614	16.14		
29	Yeongam-gun	0.01752	17.52		
30	Gurye-gun	0.02213	22.13		
31	Gimcheon-si	0.02515	25.15		
32	Jangheung-gun	0.03351	33.51	0.051657	TIER 1
33	Haenam-gun	0.04140	41.40		
34	Gangjin-gun	0.05797	57.97		
35	Sinan-gun	0.06176	61.76		
36	Goryeong-gun	0.06366	63.66		
37	Seongju-gun	0.25863	258.63		

Last row Removed this from calculations, too much of an outlier

1.4.Method 4

Table 35:

Number	Municipality	Flood waste (tons)	Population density	Density Classification	Waste/density average (tons/person/km ²)
1	Sancheong-gun	250	42	NOT DENSE	5.84
2	Hapcheon-gun	173	43		
3	Sinan-gun	2,170	52		
4	Gokseong-gun	240	57		
5	Jangheung-gun	1,182	57		
6	Boseong-gun	116	57		
7	Gurye-gun	600	61		
8	Hadong-gun	230	62		
9	Haenam-gun	2,700	62		
10	Gangjin-gun	1,900	65		
11	Jindo-gun	430	67		
12	Jangseong-gun	574	80		
13	Jangseong-gun	274	80		
14	Goryeong-gun	2,000	81		
15	Hwasun-gun	379	85		
16	Goheung-gun	1,050	87		
17	Yeongam-gun	1,000	93		
18	Namwon-si	168	108		
19	Wanju-gun	430	112		
20	Yeonggwang-gun	180	116		
21	Wando-gun	601	123		

22	Miryang-si	1,100	129		
23	Jeongeup-si	220	159		
24	Gimje-si	400	162		
25	Muan-gun	1,366	185		
26	Naju-si	80	187		
27	Ulju-gun	428	297		
28	Suncheon-si	150	308		
29	Seongju-gun	3,500	374		
30	Yangsan-si	1,600	487		
31	Hampyeong-gun	100	506		
32	Iksan-si	182	605	DENSE	2.61
33	Gunsan-si	2,365	703		
34	Gimcheon-si	3,500	1,005		
35	Pohang-si	3,703	1,133	VERY DENSE	0.60
36	Dong-gu	80	1,695		
37	Mokpo-si	530	4,348		

1.5.Method 5

Table 36: Acronyms for damaged buildings

Complete collapse	Moderate	Floor Level inundation
CC	MC	FL

Table 37: Data collected and adapted from Red Cross reports

source	Year	CC	MC	FL	Number of localities	total households affected
DREF operation Operation n°. MDRCI012	2020	279	817	1249	24	2345
DREF operation no. MDRCI015	2022	134	269	1510	10	1913
DREF n° MDRCI011	2019	72	202	2020	13	2150
DREF n° MDRCI010	2018 (adaptation)	370	983	3647	9	5000

Table 38: Database Jumia; low, mid, and high-end typical items

	Bed	chair	table	fridge	stove	tv	radio	clothes	couch
low end weight	4	n/a	5	14.5	3	5	n/a	n/a	n/a
mid end weight	18	n/a	n/a	59	4	10	n/a	n/a	n/a
high end weight	25	n/a	30	86	12	10	5	n/a	n/a
average weight	15.67	6	17.50	53.17	6.33	8.33	2	0.10	25

1.6. Models Runs

- CODE for Taiwan, Korea, 126 runs, and Bayesian updating

```
#RUN 1: LINEAR TAIWAN MODEL
```

```
library(readxl)
```

```
file_path <- "C:/Users/emman/OneDrive/Bureau/RESEARCH/new research topics/Datasets/model_four.xlsx"
```

```
flood <- read_excel(file_path, sheet="Sheet1")
```

```
flood <- flood[6:nrow(flood), ]
```

```
#standardize
```

```
# Function to standardize columns
```

```
standardize <- function(x) {
```

```
  return((x - mean(x)) / sd(x))
```

```
}
```

```
# Apply the function to each column
```

```
flood <- as.data.frame(lapply(flood, standardize))
```

```
#correlated features
```

```
cor(flood)
```

```
#model
```

```
fitdet = lm(WASTE ~ PD + FLOODED_AREA + RAINFALL, data=flood)
```

```
summary(fitdet)
```

```
## Extract R-squared value
```

```
r_squared <- summary(fitdet)$r.squared
```

```
adjusted_r_squared <- summary(fitdet)$adj.r.squared
```

```
print(r_squared)
```

```
print(adjusted_r_squared)
```

```
#vif factor for analysis
```

```
library(car)
```

```
vif(fitdet)
```

```
##coefficients
```

```
#coef(fitdet)
```

```
## Calculate the standardized residuals
```

```
#strdet <- residuals(fitdet)/sd(residuals(fitdet))
```

```
## Calculate the accuracy indices using "metrics" package
```

```
library(Metrics)
```

```
rmse(flood$WASTE, fitdet$fitted.values)
```

```
mae(flood$WASTE, fitdet$fitted.values)
```

```
##plots
```

```
library(ggplot2)
```

```
#### Create a dataframe containing the actual and fitted values
```

```
plot_data <- data.frame(Actual = flood$WASTE, Fitted = fitted(fitdet))
```

```
plot_data
```

```
#### Create a scatter plot of actual vs. fitted values
```

```
ggplot(plot_data, aes(x = Actual, y = Fitted)) +
```

```
  geom_point() +
```

```
  geom_abline(intercept = 0, slope = 1, color = "blue", linetype = "dashed") +
```

```
  labs(title = "Actual vs. Fitted Values", x = "Actual", y = "Fitted") +
```

```
  xlim(1e+06, 12e+06) +
```

```
  ylim(1e+06, 12e+06)
```

```
#### Create a scatter plot of actual vs. fitted values
```



```
ggplot(plot_data, aes(x = Actual, y = Fitted)) +  
  
  geom_point() +  
  
  geom_abline(intercept = 0, slope = 1, color = "blue", linetype = "dashed") +  
  
  labs(title = "Actual vs. Fitted Values", x = "Actual", y = "Fitted") +  
  
  xlim(-1.5, 1.5) +  
  
  ylim(-1.5, 1.5)
```

```
#####
```

```
#####
```

```
#RUN 2: EXPONENTIAL TAIWAN MODEL
```

```
library(readxl)
```

```
file_path <- "C:/Users/emman/OneDrive/Bureau/RESEARCH/new research topics/Datasets/model_four.xlsx"
```

```
flood <- read_excel(file_path, sheet="Sheet1")
```

```
flood <- flood[6:nrow(flood), ]
```

```
#correlated features
```

```
cor(flood)
```

```
#model
```

```
fitdet = lm(log(WASTE) ~ log(PD) + log(FLOODED_AREA) + log(RAINFALL), data=flood)
```

```
summary(fitdet)
```

```
## Extract R-squared value
```

```
r_squared <- summary(fitdet)$r.squared
```

```
adjusted_r_squared <- summary(fitdet)$adj.r.squared
```

```
print(r_squared)
```

```
print(adjusted_r_squared)
```

```
#vif factor for analysis
```

```
library(car)
```

```
vif(fitdet)
```

```
##coefficients
```

```
#coef(fitdet)
```

```
## Calculate the standardized residuals
```

```
#strdet <- residuals(fitdet)/sd(residuals(fitdet))
```

```
## Calculate the accuracy indices using "metrics" package
```

```
library(Metrics)
```

```
rmse(flood$WASTE, exp(fitdet$fitted.values))
```

```
mae(flood$WASTE, exp(fitdet$fitted.values))
```

```
##plots
```

```
library(ggplot2)
```

```
###Create a dataframe containing the actual and fitted values
```

```
plot_data <- data.frame(Actual = flood$WASTE, Fitted = exp(fitted(fitdet)))
```

```
plot_data
```

```
### Create a scatter plot of actual vs. fitted values
```

```
### Create a scatter plot of actual vs. fitted values
```

```
ggplot(plot_data, aes(x = Actual, y = Fitted)) +  
  
  geom_point() +  
  
  geom_abline(intercept = 0, slope = 1, color = "blue", linetype = "dashed") +  
  
  labs(title = "Actual vs. Fitted Values", x = "Actual", y = "Fitted") +  
  
  xlim(1e+06, 12e+06) +  
  
  ylim(1e+06, 12e+06)  
  
#####  
  
#####
```

```
#RUN 3: LINEAR KOREA MODEL WITH 6 FEATURES
```

```
flood <- read.csv("regression_run.csv")
```

```
flood <- flood[6:nrow(flood), ]
```

```
fitdet = lm(WASTE ~ AREA + GRDP + PD + UR + WIND  
            + RTOTAL, data = flood)
```

```
## Extract R-squared value
```

```
r_squared <- summary(fitdet)$r.squared
```

```
adjusted_r_squared <- summary(fitdet)$adj.r.squared
```

```
print(r_squared)
```

```
print(adjusted_r_squared)
```

```
# Calculate the accuracy indices using "metrics" package
```

```
## Calculate the accuracy indices using "metrics" package
```

```
library(Metrics)
```

```
rmse(flood$WASTE, fitdet$fitted.values)
```

```
mae(flood$WASTE, fitdet$fitted.values)
```

```
##plots
```

```
library(ggplot2)
```

```
###Create a dataframe containing the actual and fitted values
```

```
plot_data <- data.frame(Actual = flood$WASTE, Fitted = fitted(fitdet))
```

```
plot_data
```

```
### Create a scatter plot of actual vs. fitted values
```

```
ggplot(plot_data, aes(x = Actual, y = Fitted)) +
```

```
geom_point() +  
geom_abline(intercept = 0, slope = 1, color = "blue", linetype = "dashed") +  
labs(title = "Actual vs. Fitted Values", x = "Actual", y = "Fitted") +  
xlim(1e+06, 12e+06) +  
ylim(1e+06, 12e+06)
```

```
#####  
#####
```

```
#RUN 4: EXPONENTIAL KOREA MODEL WITH 6 FEATURES
```



```
flood <- read.csv("regression_run.csv")
```

```
flood <- flood[6:nrow(flood), ]
```

```
fitdet = lm(log(WASTE) ~ log(AREA) + log(GRDP) + log(PD) + log(UR) + log(WIND)  
            + log(RTOTAL) + log(ALTAVG) + log(ALTMAX), data = flood)
```

```
## Extract R-squared value
```

```
r_squared <- summary(fitdet)$r.squared
```

```
adjusted_r_squared <- summary(fitdet)$adj.r.squared
```

```
print(r_squared)
```

```
print(adjusted_r_squared)
```

```
# Calculate the accuracy indices using "metrics" package
```

```
## Calculate the accuracy indices using "metrics" package
```

```
library(Metrics)
```

```
rmse(flood$WASTE, exp(fitdet$fitted.values))
```

```
mae(flood$WASTE, exp(fitdet$fitted.values))
```

```
##plots
```

```
library(ggplot2)
```

```
###Create a dataframe containing the actual and fitted values
```

```
plot_data <- data.frame(Actual = flood$WASTE, Fitted = exp(fitted(fitdet)))
```

```
### Create a scatter plot of actual vs. fitted values
```

```
ggplot(plot_data, aes(x = Actual, y = Fitted)) +
```

```
  geom_point() +
```

```
geom_abline(intercept = 0, slope = 1, color = "blue", linetype = "dashed") +  
labs(title = "Actual vs. Fitted Values", x = "Actual", y = "Fitted") +  
xlim(1e+06, 12e+06) +  
ylim(1e+06, 12e+06)
```

```
#####
```

```
#####
```

```
#RUN 5: 4-FEATURES MODEL WITH DURATION and ALTITUDES INCLUDED
```

```
flood <- read.csv("regression_run.csv")
```

```
flood <- flood[6:nrow(flood), ]
```

```
#combinations of what we want to loop through
```

```
## Define the features first
```

```
features <- c("DURATION", "AREA", "GRDP", "PD", "UR", "WIND", "RTOTAL", "ALTMIN", "ALTAVG", "ALTMAX")
```

```
##now the combinations
```

```
combinations <- combn(features, 4, simplify = FALSE)
```

```
# Initialize a list to store adjusted R-squared values and the corresponding models
```

```
results <- list()
```

```
# Loop over each combination and fit the regression model

for (comb in combinations) {

  # Create the formula

  formula <- as.formula(paste("WASTE ~", paste(comb, collapse = " + ")))

  # Fit the linear model

  model <- lm(formula, data = flood)

  # Extract the adjusted R-squared value

  adj_r_squared <- summary(model)$adj.r.squared

  # Store the combination of features and the adjusted R-squared value

  results[[paste(comb, collapse = " + ")] <- adj_r_squared

}
```

```
# Convert the results to a data frame for easier sorting and inspection
```

```
results_df <- data.frame(
```

```
  Features = names(results),
```

```
  Adjusted_R_Squared = unlist(results)
```

```
)
```

```
# Sort the results by adjusted R-squared in descending order
```

```
sorted_results <- results_df[order(-results_df$Adjusted_R_Squared), ]
```

```
# Display the top results
```

```
print(sorted_results)
```

```
# Calculate the accuracy indices using "metrics" package
```

```
#####INSERT loop HERE FOR TOP 5 SCORING MODELS
```

```
library(Metrics)
```

```
rmse(flood$WASTE, fitdet$fitted.values)
```

```
mae(flood$WASTE, fitdet$fitted.values)
```

```
##plots
```

```
#####INSERT loop HERE FOR TOP 5 SCORING MODELS
```

```
library(ggplot2)
```

```
###Create a dataframe containing the actual and fitted values
```

```
plot_data <- data.frame(Actual = flood$WASTE, Fitted = exp(fitted(fitdet)))
```

```
### Create a scatter plot of actual vs. fitted values
```

```
ggplot(plot_data, aes(x = Actual, y = Fitted)) +
```

```
  geom_point() +
```

```
  geom_abline(intercept = 0, slope = 1, color = "blue", linetype = "dashed") +
```

```
  labs(title = "Actual vs. Fitted Values", x = "Actual", y = "Fitted") +
```

```
  xlim(1e+06, 12e+06) +
```

```
ylim(1e+06, 12e+06)
```

```
#####
```

```
#####
```

```
#RUN 6: 3-FEATURES MODEL WITH DURATION INCLUDED
```

```
flood <- read.csv('regression_run.csv')
```

```
# Define the features
```

```
features <- c("DURATION", "AREA", "GRDP", "PD", "UR", "WIND", "RTOTAL", "ALTMIN", "ALTAVG", "ALTMAX")
```



```
# Generate all combinations of 4 features out of 9
```

```
combinations <- combn(features, 3, simplify = FALSE)
```

```
# Initialize a list to store adjusted R-squared values and the corresponding models
```

```
results <- list()
```

```
# Loop over each combination and fit the regression model
```

```
for (comb in combinations) {
```

```
  # Create the formula
```

```
  formula <- as.formula(paste("WASTE ~", paste(comb, collapse = " + ")))
```

```
  # Fit the linear model
```

```
  model <- lm(formula, data = flood)
```

```
  # Extract the adjusted R-squared value
```

```
adj_r_squared <- summary(model)$adj.r.squared
```

```
# Store the combination of features and the adjusted R-squared value
```

```
results[[paste(comb, collapse = " + ")] <- adj_r_squared
```

```
}
```

```
# Convert the results to a data frame for easier sorting and inspection
```

```
results_df <- data.frame(
```

```
  Features = names(results),
```

```
  Adjusted_R_Squared = unlist(results)
```

```
)
```

```
# Sort the results by adjusted R-squared in descending order
```

```
sorted_results <- results_df[order(-results_df$Adjusted_R_Squared), ]
```

```
# Display the top results
```

```
print(sorted_results)
```

```
# Calculate the accuracy indices using "metrics" package
```

```
library(Metrics)
```

```
rmse(flood$WASTE, fitdet$fitted.values)
```

```
mae(flood$WASTE, fitdet$fitted.values)
```

```
##plots
```

```
library(ggplot2)
```

```
###Create a dataframe containing the actual and fitted values
```

```
plot_data <- data.frame(Actual = flood$WASTE, Fitted = exp(fitted(fitdet)))
```

```
### Create a scatter plot of actual vs. fitted values
```

```
ggplot(plot_data, aes(x = Actual, y = Fitted)) +  
  
  geom_point() +  
  
  geom_abline(intercept = 0, slope = 1, color = "blue", linetype = "dashed") +  
  
  labs(title = "Actual vs. Fitted Values", x = "Actual", y = "Fitted") +  
  
  xlim(1e+06, 12e+06) +  
  
  ylim(1e+06, 12e+06)
```

Table 39: Runs with 3-feature models

Rank	The 3 Features	Adjusted R ²
1	PD + WIND + RTOTAL	0.742
2	PD + WIND + DURATION	0.703
3	PD + DURATION + RTOTAL	0.693
4	DURATION + WIND + AREA	0.633
5	DURATION + AREA + RTOTAL	0.633

2. Appendix B : EIA

How the mass of each construction material was found: “The densities being known, in addition to the 8m³ volume of one slum house (construction materials only), it was assumed a 80/20/19 % split for building/binding/roof in terms of volume repartition. The remaining 1% was attributed to the liquids and additives. From this, the mass of each item was estimated”.

Table 40: Percentage of each metal from all metals

METALS	Mass (kg)	
Copper	11724	1.62%
Lead	115	0.02%
Steel	16507	2.28%
Mercury	0.1184	0.00%
Nickel	2863	0.40%
Zinc	2297	0.32%
Aluminium	689199	95.36%
TOTAL	722705	100%